

Honeywell

Honeywell Hi-Spec Solutions

ProMenu User Guide

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About This Document

This document describes about ProMenus and how to create them.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
GUSPro Installation Guide	GP05-110
ProTrend User Guide	GP21-110
GUSPro Software Change Notice	GPro110-SCN
GUSPro Specification and Technical Data	GP03-110
Display Builder User's Guide	GU23

Support and Other Contacts

Document Title	Doc ID
Display Scripting User's Guide	GU24
Display Authoring Tutorial	GU07
SafeView User's Guide	GU27

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.

Symbol Definitions

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1. Introduction

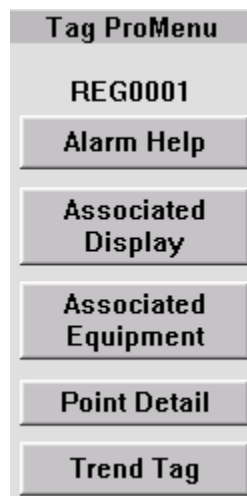
1.1 Overview

ProMenu provides context-sensitive navigation from within a GUS graphic display directly to the applications and, as well as to other applications, documents and GUS displays.

ProMenu Graphics

A ProMenu graphic is a user defined GUS graphic that contains a series of Link buttons associated with a given tag, equipment item or equipment group. Each Link button provides access to a predefined document, display or application that is related to the selected tag or equipment.

Figure 1 Example Tag ProMenu



By grouping the Link buttons together within a single display, the user is provided with easy access to all of the functions and information that might be needed for a given tag, equipment item or group of equipment items. For example, users could access trends, events, alerts, startup procedures and shutdown procedures for a piece of equipment all from the same place.

1. Introduction

1.1. Overview

In addition to the Link buttons, each ProMenu identifies the tag, equipment item or equipment group that it was called for. Any errors that result from user interactions with the ProMenu itself are also displayed.

Each site requires its own set of ProMenus. At minimum, a separate ProMenu graphic is required for equipment and for tags. In many cases, different ProMenus are needed for different types of equipment and tags.

Tag ProMenu Example

Figure 1 shows an example ProMenu for the selected tag, REG0001. The Link buttons on the Tag ProMenu call up documents, displays and applications specific to that tag. The example includes the following Link buttons:

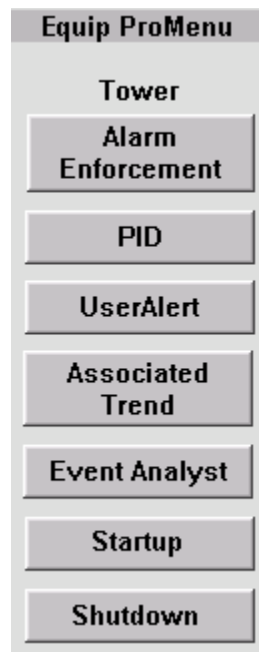
Alarm Help	Calls up Alarm Help information for the selected tag.
Associated Display	Calls up the Associated display configured for the selected tag.
Associated Equipment	Calls up the Equipment ProMenu that the selected tag is associated with.
Point Detail	Calls up the Point Detail display for the selected tag.
Trend Tag	Calls up the ProTrend Tool and adds the selected tag's Process Variable (PV) to the trend.
UserAlert	Calls up the UserAlert Monitor display showing alerts for the current console.

If a faceplate or change zone display is being used to make process changes, the ProMenu Link buttons may be placed on the display. In this case the Link buttons function the same as they would on the ProMenu graphic for the same tag.

Equipment ProMenu Example

Figure 2 shows a ProMenu for a selected equipment item called Tower. The ProMenu call up button on the Tower invokes the appropriate Equipment ProMenu.

Figure 2 Example Equipment ProMenu



The ProMenu provides links to information and functions associated with the Tower. The following Link buttons have been defined for this ProMenu:

Alarm Enforcement	Calls up the ACM Alarm Enforcer display for the selected equipment.
PID	Calls up a custom display showing the P&ID for the selected equipment.
UserAlert	Calls up the UserAlert Monitor display, showing alerts for the selected equipment.
Associated Trend	Calls up the prebuilt Trend display related to the selected equipment.

1. Introduction

1.1. Overview

Event Analyst	Calls up the Event Analyst display, showing events and trends related to the selected equipment, based on the associated User File.
Startup	Calls up the startup procedures document for the selected equipment.
Shutdown	Calls up the shutdown procedures document for the selected equipment.

The implementer may also define custom Link buttons to access other applications and documents, such as work orders and maintenance history.

GUSPro Consoles

As an aid for managing the operator workspace, ProMenus also control where the requested document, display or application is actually invoked. To do this, one or more *GUSPro* consoles are defined.

A *GUSPro* console is a virtual console that consists of a group of nodes. In general, each *GUSPro* console includes the nodes in the operator view and the GUS stations that are part of a single TPN console, along with any supporting nodes.

As part of *GUSPro* console configuration every application, document and display that can be launched from a ProMenu must be identified. The nodes where each application, document and display is expected to run is also configured. This information is stored in the Console database.

When a ProMenu button is selected, the local Console database indicates whether the requested application, document or display can be launched locally, or, if not, which node to launch it on.

1.2 Using ProMenus

Software Components

To use a ProMenu, the software components, shown in the following diagram and described in Table 1, take part in the operation.

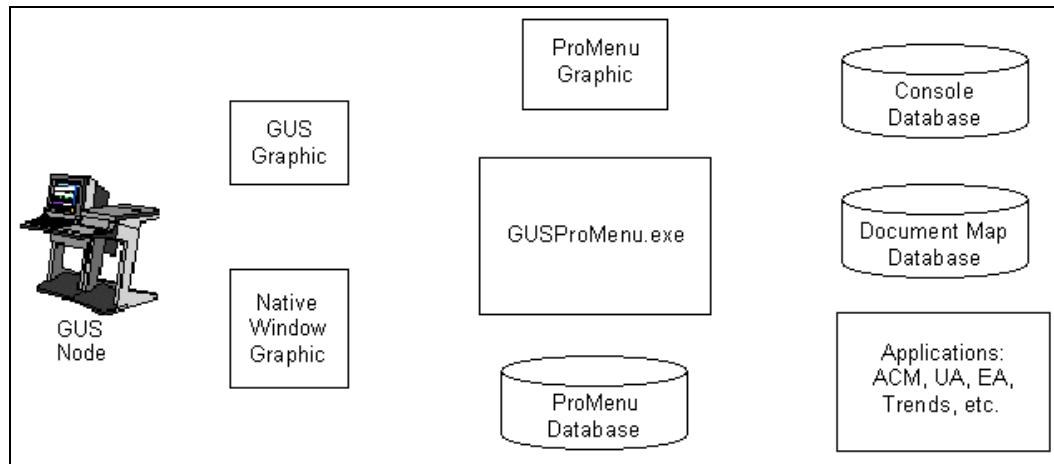


Table 1 ProMenu Software Components

Component	Description
GUS Graphics	The user's own GUS graphic displays are customized to include buttons or shortcut menus that provide access to ProMenu graphics.
Native Window Graphics	In a manner similar to GUS graphics, the Universal Schematics, which run in the Native Window of a GUS, are also customized to provide access to ProMenu graphics.
ProMenu Graphics	ProMenu graphics are created for tags, equipment and equipment groups to suit the user's needs. The Link buttons in the ProMenu graphics are used to request the associated applications, documents and displays.

1. Introduction

1.2. Using ProMenus

Component	Description
ProMenu Application (GUSProMenu.exe)	The ProMenu application is the engine that drives ProMenu operations. When a ProMenu is requested, this application determines which ProMenu graphic is invoked for a given tag or equipment name. When a Link button in a ProMenu graphic is selected, the ProMenu application determines which node the requested application is to be invoked on and then launches the application on the appropriate computer.
ProMenu Database	The ProMenu database defines the mapping between the available ProMenu graphics and the tag, equipment and equipment group names, based on a series of rules. When a ProMenu is requested, the ProMenu application searches the contents of the ProMenu database for the ProMenu graphic associated with the selected tag, equipment or equipment group name.

Component	Description
Document Map Databases	The Document Map databases define the mapping between specific documents (for example, startup procedures) and the tag, equipment or equipment group names, based on a series of rules. A separate Document Map database is created for each type of document that can be launched from a ProMenu graphic. When a ProMenu Link button for a document is selected, the ProMenu application searches the contents of the appropriate Document Map database for the document associated with the current tag, equipment or equipment group name. The ProMenu application can deal with two types of documents, documents that follow a fixed, ProMenu naming convention and documents that do not. If all of the documents that are to be launched via ProMenu graphics follow the ProMenu naming convention, then Document Maps (and Document Map databases) are not required. Document Maps are provided to allow users to take advantage of their existing documents and to make them accessible from the ProMenus.
Console Database	The Console databases specify which applications, documents and displays, may be invoked on each node in a GUSPro console. A separate Console database is created for each GUSPro console in the system. Once the ProMenu application has determined which document, display or application is to be invoked, the ProMenu application searches the contents of the local Console database to determine whether the requested application, document or display can be launched locally, or, if not, which node to launch it on.

1. Introduction

1.2. Using ProMenus

Component	Description
Applications	The implementer determines the set of applications, displays or documents that can be launched from a ProMenu graphic Link button. In general, the applications could include any of the following: • Applications – Alarm Configuration Manager Alarm Enforcer – Event Analyst – UserAlert Monitor – ProTrend Tool – Alarm Help • Standard GUS and Native Window Displays – Point Details for tags – Associated Displays – Faceplate Displays • Miscellaneous Documents (which use the ProMenu document naming convention or which have a Document Map) • Custom Applications (for example, custom trend displays, P&ID drawings, and other GUS graphics) • Other Associated ProMenu Graphics
ProMenuStartup.exe	This application has a shortcut in the Startup folder and runs whenever a user logs on to the node. Its role is to create an instance of the GUSProMenu application that keeps running until the user logs off (or until the ProMenuStartup application is manually stopped using the task manager). This allows the properties of the GUSProMenu application to be persistent.

Component	Description
InitializeGUSProSettings.exe	This application has a shortcut in the Startup folder and runs whenever a user logs on to the node. Its role is to initialize the registry settings used by the ProMenu and Trend Template graphics. These settings include colors for all of the GUSPro graphics as well as time spans and so on for the trend templates. The master copy of the registry settings is located in HKEY_LOCAL_MACHINE. When the InitializeGUSProSettings application runs it copies the contents of the \\BasicScript Program Settings\\Software\\GUSPro key from HKEY_LOCAL_MACHINE to HKEY_CURRENT_USER. The application resides in the %GUSProDir%\\Base directory and can also be run with a command line. It has one command line argument: "-r". This argument causes any settings in HKEY_CURRENT_USER to be replaced, otherwise the GUSPro key in HKEY_CURRENT_USER is written only if it does not already exist. By default the shortcut in the Startup folder uses the -r argument. If this default is used any changes to the registry values must be made in HKEY_LOCAL_MACHINE.

1. Introduction

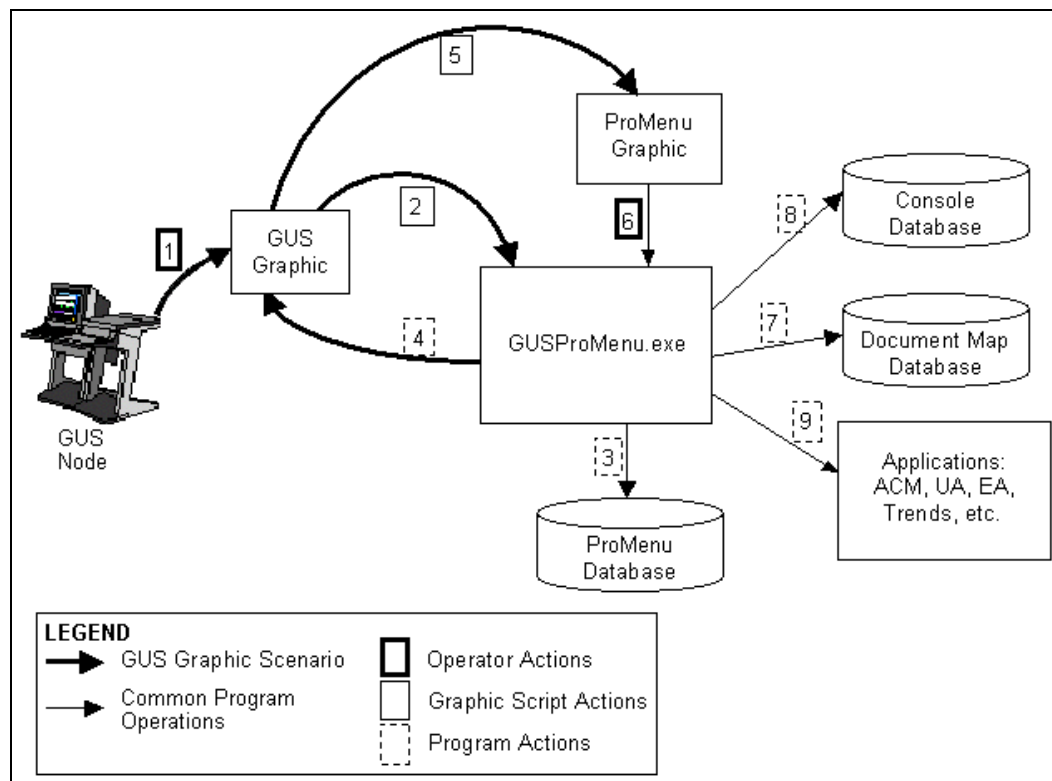
1.2. Using ProMenus

Operating Scenarios

Since ProMenus can be launched from either a GUS graphic or a Native Window schematic, this results in the two operating scenarios described below.

GUS Graphic Operating Scenario

The following figure shows how ProMenus are accessed and used from a GUS graphic display.



In general, there are three different ways for actions to occur. Operator actions are actions that result from the interaction between the operator and the GUS graphic. Graphic script actions occur as a result of coding on the GUS and ProMenu graphics. Program actions occur within the ProMenu application, GUSProMenu.exe.

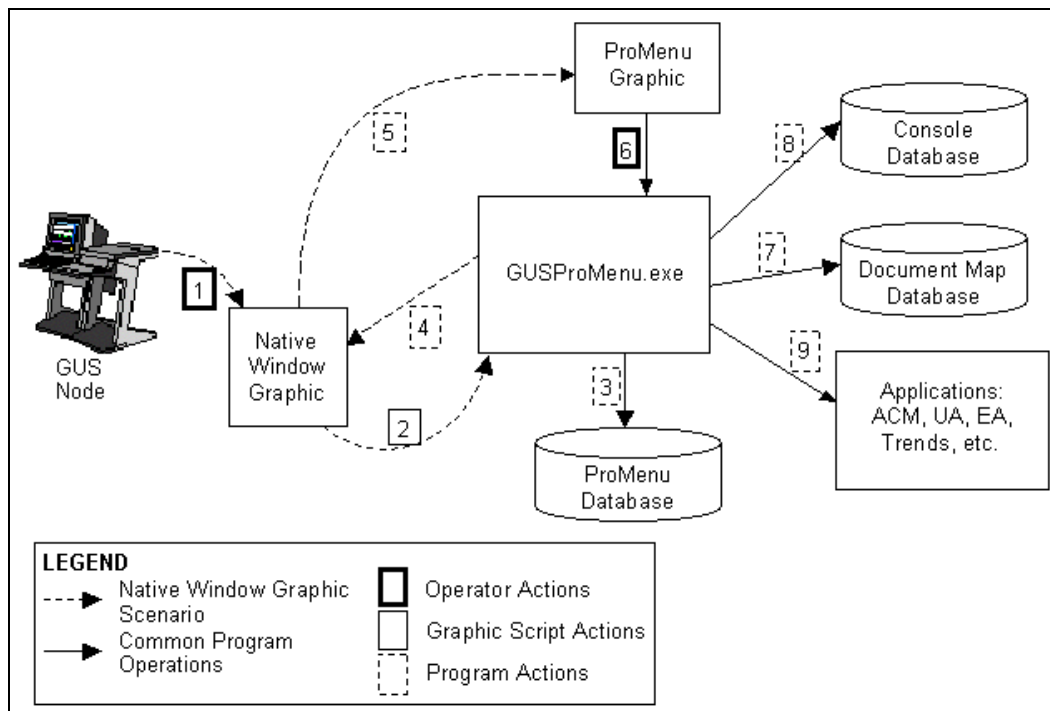
Step	Action
1	From a GUS graphic display, the operator touches a ProMenu target associated with a specific tag, equipment item or equipment group.
2	A script on the target calls the GetProMenu method on the GUSProMenu.Operations interface (in the GUSProMenu.exe object).
3	The GetProMenu method accesses the ProMenu database to determine which ProMenu is to be invoked, based on the selected tag or equipment name.
4	The name of the ProMenu graphic is returned to the GUS graphic display.
5	A script on the target invokes the ProMenu graphic returned by the GetProMenu method. NOTE: The graphic must be stored in a folder that is part of the GUS search path.
6	The operator chooses a button on the ProMenu graphic. This runs a script that runs the QuickAppRun method.
7	For instances where the application is looking for a document map, the application checks the Document Map database to determine which document must be called up.
8	Within the ProMenu application the GetApplicationHost command is run to determine where the command must be run. The Console database is accessed to determine which node the requested application is to be launched on.
9	The RunApplication command is called by the ProMenu application on the appropriate node.

1. Introduction

1.2. Using ProMenus

Native Window Graphic Operating Scenario

The following figure shows how ProMenus are accessed and used from a Native Window graphic display. (NOTE: ProMenus are always GUS graphics, even if they are invoked from Native Window.)



As in the GUS Graphic Scenario, there are three different ways for actions to occur. Operator actions are actions that result from the interaction between the operator and the Native Window graphic. Graphic script actions occur as a result of coding on the Native Window and ProMenu graphics. Program actions occur within the ProMenu application, GUSProMenu.exe.

Step	Action
1	From a Native Window graphic display, the operator touches a ProMenu target associated with a specific tag, equipment item or equipment group.
2	The target contains the GP_Menu actor script, which calls the GetProMenu method on the GUSProMenu.exe object.
3	GetProMenu method accesses the ProMenu database to determine which ProMenu is to be invoked, based on the selected tag or equipment name.
4	The GetProMenu method returns the name of the graphic to be invoked to the GP_Menu actor.
5	The GP_Menu actor invokes the ProMenu Graphic. NOTE: The graphic must be stored in a folder that is part of the GUS search path.
6	The operator chooses a button on the ProMenu graphic. This runs a script that runs the QuickAppRun method.
7	For instances where the application is looking for a document map, the application checks the Document Map database to determine which document must be called up.
8	Within the ProMenu application the GetApplicationHost command is run to determine where the command must be run. The Console database is accessed to determine which node the requested application is to be launched on.
9	The RunApplication command is called by the ProMenu application on the appropriate node.

1. Introduction

1.3. Background Information

1.3 Background Information

Before proceeding with creating ProMenus, some additional background information is necessary to understand how ProMenus work and how they are configured.

ProMenu Implementation

ProMenus are implemented as GUS displays with Link button subpictures that invoke applications, documents, or GUS displays that are related to the currently selected context. Each ProMenu is invoked from a tag or equipment object within a GUS display or within a Universal Station schematic running in the Native Window of a GUS.

The implementer defines the mechanisms within a display that call up ProMenus. Typically, a Tag ProMenu is called up by right clicking on a tag object within a GUS display. A tag object is an object that represents a single DCS point (for example, valve, pump, or value display). If a faceplate or change zone display is being used for making process changes, the Tag ProMenu may be called up from a button placed in the display. The Tag ProMenu Link buttons may also be put directly in the displays.

Similarly, Equipment ProMenus are called up by clicking on equipment objects within a GUS display or on a button associated with those objects. For example, a tower could be set up as an equipment object.

ProMenus may also be called up from an object within Universal Station schematics running in the Native Window of a GUS. Actor scripts are provided that the implementer can put into existing schematics to call up the ProMenus. The ProMenus and Link buttons function the same, regardless of whether they are called up from GUS displays or Universal Station schematics.

ProMenu Database

Although generally thousands of tag and equipment groups are defined, there are typically only a few ProMenus created. To support this relationship, of many tags to one ProMenu, ProMenu rules are defined. ProMenu rules are used to associate tag or equipment names with ProMenu graphics based on characters in the tag or equipment name. For example to specify all tags that contain the string LIC, the ProMenu rule *LIC* could be used. Tags and equipment are separated to allow for a different default ProMenu for tags and for equipment.

ProMenu rules can contain wildcard characters (? = any single character, * = any number of characters). They are used to define which ProMenu is to be called up when an equipment or tag name is passed to the ProMenu application.

The ProMenu Configuration utility is used to configure the ProMenu rules for tags and equipment. It is also used to assign each ProMenu rule to a ProMenu graphic. Each ProMenu graphic can be associated with many rules. Once this information has been configured, a ProMenu database file is generated. The ProMenu database defines the mapping of ProMenu rules to ProMenu graphics. (See the ProMenu Database Configuration section for a detailed description of ProMenu database configuration.)

Naming Conventions

The following ProMenu naming conventions are defined for the standard documents and applications arguments used by ProMenus. In all cases the <name> entry is the equipment item, equipment group or tag that the file corresponds to.

1. Introduction

1.3. Background Information

File Type	File Name Format	Description
GUS Graphics	<DisplayType>_<name>.pct	Used to identify a group of GUS graphics of the same type (for example, P&IDs or trend displays), which can be launched in a context-sensitive manner from a ProMenu, where: DisplayType = The name used to identify the set of GUS graphic displays of the same type and which is configured in the Console database. For example, a GUS graphic for a trend associated with an equipment group called "tower", must be labeled "TRENDISP_TOWER.pct". In this case the string "TRENDISP" must be defined in the console as a GUS display type.
ProMenu Document Naming Conventions	<PrefixString>_<name>.<DocumentExtension>	Used to identify a group of documents of the same type (for example, startup or shutdown procedures), which can be launched in a context-sensitive manner from a ProMenu, where: PrefixString = The name used to identify the set of documents of the same type and which is configured in the Console database. DocumentExtension = The file name extension for the set of documents. For example, the startup procedure for an equipment group called "boiler", must be labeled "STARTUP_BOILER.doc" In this case the prefix string "STARTUP" must be defined in the console as a ProMenu Naming Convention Doc. document type.
Event Analyst User Files	EA_<name>.txt	Used to identify the Event Analyst User Files that can be launched in a context-sensitive manner from a ProMenu for a specific equipment item.

Document Map Databases

For those documents that do not use the ProMenu naming convention for documents, Document Maps can be built that specify which document is opened for a given tag, equipment or equipment group name. Much like ProMenu rules, Document Map rules are used to match tag, equipment or equipment group names to documents.

The Document Map databases define the mapping of Document Map rules to documents. One Document Map database must exist for every document type being mapped. For example, there may be a startup procedures Document Map and a shutdown procedures Document Map.

The Document Map Configuration utility is used to create and configure the Document Map databases. (See the Document Map Configuration section for a detailed description of Document Map configuration.)

If all of the documents to be launched via the ProMenus are named using the standard ProMenu naming convention, then Document Map databases are not required.

Console Database

As mentioned previously, the Console database specifies which applications, documents and displays may be invoked on each node in a *GUSPro* console. A separate Console database is created for each *GUSPro* console in the system. The Console Configuration utility is used to create and configure the Console databases. (See the Console Configuration section for a detailed description of console configuration.)

Each *GUSPro* console is matched with a corresponding ACM console. This makes it possible for the equipment and equipment groups that are configured in ACM, to be used to carry out *GUSPro* configuration. UserAlert also takes advantage of the ACM console defined in the local *GUSPro* console to carry out various *GUSPro*-specific tasks.

1. Introduction

1.4. Creating ProMenus

1.4 Creating ProMenus

The task of creating ProMenus can be broken down into the following phases:

1. Planning (see ProMenu Planning)
2. Configuration (see ProMenu Configuration, ProMenu Database Configuration, Document Map Configuration, Console Configuration)
3. Implementation (see ProMenu Implementation, ProMenu Buttons Reference, Building a ProMenu Display, SafeView Configuration, Graphic Scripting for ProMenus, ProMenu Application Programming Interface)

Each of these phases is described separately in the identified sections.

ProMenu Development Toolkit

The ProMenu Development Toolkit provides the components and tools needed to create custom ProMenus for a given solution. The components are:

- Link Buttons and Subpictures
- ProMenu Template
- ProMenu Configuration Utility
- Document Map Configuration Utility
- Console Configuration Utility
- SafeView Templates
- Component Explorer Library for the Link Buttons

Link Buttons and Subpictures

Link buttons are GUS embedded displays that invoke specific documents, applications and displays that are associated with the selected tag or equipment. The following are the Link buttons and subpictures are provided in the ProMenu toolkit.

Common Subpictures	Equipment ProMenu Subpictures	Tag ProMenu Subpictures
• ProMenu Title • ProMenu Document Naming Convention Button • Document Map Button • UserAlert Button • Custom Application Button	• Associated Trend Button • Event Analyst Button • Alarm Enforcement Button	• Alarm Help Button • Associated Display Button • Associated Equipment Menu Button • Point Detail Button • Trend Tag Detail Button

These subpictures are added to ProMenu templates by the implementer to create custom ProMenus for tag and equipment types. They invoke the appropriate documents, displays and applications, based on the selected tag or equipment name, which is stored in a global variable. (See ProMenu Buttons Reference section for a detailed description of the ProMenu Link buttons and subpictures.)

1. Introduction

1.4. Creating ProMenus

2. ProMenu Planning

2.1 Planning Tasks

The first step in creating ProMenus, is to determine which ProMenus are needed along with the tags, equipment and equipment groups they are going to be apply to.



TIP

Be sure that you work closely with the plant engineering and operations personnel to ensure that the ProMenus meet the existing facility's operating requirements.

The following ProMenu planning tasks must be completed before you proceed with ProMenu configuration or implementation:

Task	Description
1	Identify the set of Tag ProMenu graphics that need to be created, along with the set of tags that uses each Tag ProMenu.
2	Based on the names of the tags that access each Tag ProMenu, define the set of tag name rules that causes the correct ProMenu to be launched for each tag.
3	Identify the set of Equipment ProMenu graphics that need to be created, along with the set of equipment or equipment group names that uses each Equipment ProMenu.
	TIP Wherever possible use the equipment groups defined in the ACM equipment hierarchy. (See the <i>ACM Alarm Manager User Guide</i> for details.)
4	Based on the equipment names that access each Equipment ProMenu, define the set of equipment name rules that causes the correct ProMenu to be launched for each equipment item or group.

2. ProMenu Planning

2.1. Planning Tasks

Task	Description
	TIP 1. As an aid to configuring the ProMenu database, the set of equipment and equipment group names can be collected directly from ACM. If ACM is not available, the equipment names can also be collected in a Microsoft Excel workbook. 2. Similarly, the set of tag names can be collected in a Microsoft Excel workbook. Be sure to use a separate worksheet (or workbook) for tag names, and for equipment and equipment group names. 3. An Excel worksheet containing the set of ProMenu graphic file names and the tag or equipment rules that apply to each one can also be very helpful in ProMenu database configuration. Use a separate worksheet for Tag ProMenus and Equipment ProMenus. 4. See the Importing Lists from Excel section for a description of the Excel worksheet layouts that must be used to import tag and equipment names, and ProMenu rule lists into the ProMenu Configuration utility.
5	Identify the applications, displays and documents (that is, Link buttons) that are to be accessible from each ProMenu.
6 	For any documents that do not follow the ProMenu document naming convention, define the set of tag or equipment name rules that are used to map the documents to the appropriate tags or equipment. TIP Like the ProMenu rules lists, putting the Document Map rules in an Excel worksheet is very helpful for Document Map database configuration. See the Importing Lists from Excel section for a description of the Excel worksheet layouts that must be used to import Document Map rule lists into the Document Map Configuration utility.
7	Identify the set of nodes that are going to belong to each GUSPro console. The machine name or IP address is required for each node.
8	For each node in each GUSPro console, identify the applications, displays and documents that exist and are permitted to be run on that node.
9	For each application, display and document that can be accessed from a ProMenu, decide which node (or nodes) in the GUSPro console that it can, or must, be launched on. Be sure to consider how the operator workspace is to be managed and the actual locations where the applications, displays and documents are installed.

Task	Description
10	For each <i>GUSPro</i> console, identify the corresponding ACM console. In addition, identify the ACM control system name.
11	Identify the SafeView configuration that is going to be used for the ProMenus on each GUS node. (See SafeView Templates Provided for a description of the SafeView templates provided.)
12	Decide how the Tag and Equipment ProMenus are to be accessed (right-click or left-click) from the existing GUS or Native Window graphics.
13	Based on the site's existing replication scenario, decide where the new Tag and Equipment ProMenu graphic files are to be stored. The same directory path names must be used on every node, where the ProMenu graphics are expected to run. Similarly, decide where the ProMenu database, Document Map database (if required) and Console database files are to be stored. The same directory path names must be used on every node, where the ProMenu application is installed. To simplify this task, a set of default directory path names are identified in the <i>GUSPro</i> Settings display. These names can be used as is or altered to suit site requirements. (See <i>GUSPro</i> Settings Display for details.)

2. ProMenu Planning

2.1. Planning Tasks

3. ProMenu Configuration

3.1 Configuration Tasks

Once the ProMenu planning tasks have been completed, the following ProMenu configuration tasks can be carried out:

Task	Description	Reference
1	Update the file storage location settings for the Equipment and Tag ProMenu graphic files, ProMenu database, Document Map databases (if required) and Console databases, in the GUSPro Settings display. On every node where the ProMenu graphics are expected to run, set up the Equipment and Tag ProMenu Graphics directory paths specified in the GUSPro Settings display. On every node where the ProMenu application is installed, set up the ProMenu database, Document Map database (if required) and Console database directory paths specified in the GUSPro Settings display.	see GUSPro Settings Display section
2	Configure the ProMenu database using the ProMenu Configuration utility. Set up the relationships between the ProMenu graphics and the tags or equipment they are to be launched for, based on the rules identified in the ProMenu planning phase.  TIP It is helpful, but not required, if the Tag and Equipment ProMenu graphic files have been created before the ProMenu database is configured. The ProMenu Configuration utility can automatically collect the names of the existing graphic files, if they have been stored in the assigned Tag and Equipment ProMenu graphic storage directories.	See ProMenu Database Configuration section

3. ProMenu Configuration

3.1. Configuration Tasks

Task	Description	Reference
3	If required, configure the Document Map databases using the Document Map Configuration utility. Set up the relationships between the documents and the tags or equipment they are to be launched for, based on the rules identified in the ProMenu planning phase. As stated earlier, Document Map databases are only required for documents that are going to be accessed from the ProMenu graphics and that do not follow the ProMenu document naming convention.	see Document Map Configuration section
4	Configure the Console database for each GUSPro console using the Console Configuration utility. This involves the following operations: a) Assign the nodes to each GUSPro console. b) Configure every application, document and GUS graphic display type that can be launched from the ProMenus. (NOTE: The standard client applications: ACM Alarm Enforcer, Event Analyst, UserAlert and ProTrend Tool, are configured for use automatically.) c) Identify the applications, documents and displays that can be run on each node. d) Identify the nodes where each application, document and display is to be launched, when it is requested from a ProMenu. This includes specifying the node selection order in the database for each application, document and display. e) Identify the ACM control system name and assigning each GUSPro console to its corresponding ACM console.	See Console Configuration section
5	Replicate the resultant database and configuration files to each node as directed.	



TIP

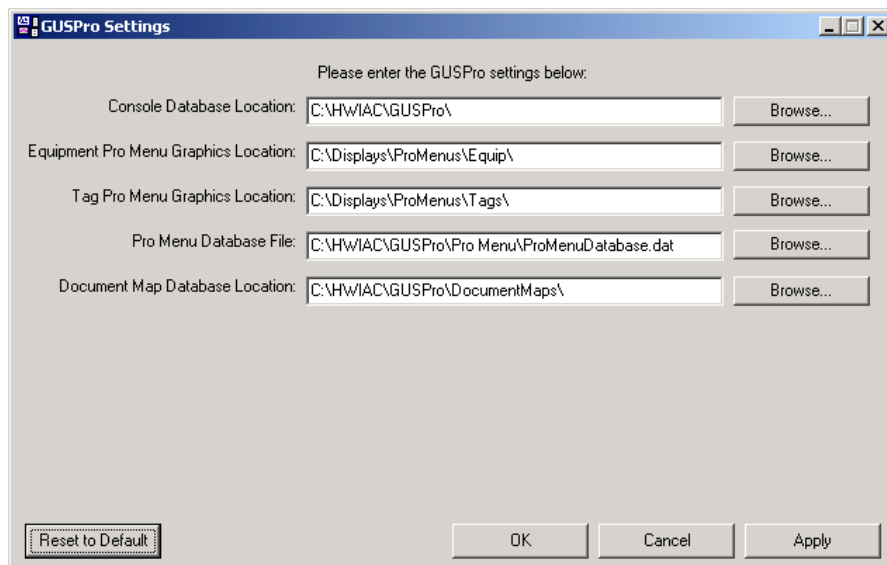
For consistency, all of the ProMenu configuration tasks must be carried out on one of the GUSPro Client computers. By using the same computer for all of the configuration tasks, duplication of effort and configuration errors can be avoided.

3.2 GUSPro Settings Display

Purpose

The GUSPro Settings display (Figure 3) is used to change the file storage location settings used by the ProMenu application.

Figure 3 GUSPro Settings Display



TIP

Most of the settings in the GUSPro Settings display can also be changed in the other Configuration utilities associated with ProMenu. Any changes made here are automatically reflected in the associated Configuration utility and vice versa.

Access

To access the GUSPro Settings display, be sure that you are logged on using an account with local administrator privileges. Launch the GUSPro Configuration utility from the Windows **Start** menu by selecting **Programs > Honeywell OPS > GUSPro > GUSPro Configuration**. In the GUSPro Configuration utility, click the **GUSPro Settings** button.

3. ProMenu Configuration

3.2. GUSPro Settings Display

Functions and Features

The GUSPro Settings display contains a series of text boxes defining the file storage locations for the ProMenu files.

Table 2 ProMenu File Storage Settings

Setting	Description	Default
Console Database Location	Defines the storage directory for the Console database files. The Console database files are created and maintained using the Console Configuration utility, described in Console Configuration Utility section.	C:\HWIAC\GUSPro\
Equipment ProMenu Graphics Location	Defines where the ProMenu GUS graphics associated with equipment are stored. (see ATTENTION box following this table)	C:\Displays\ProMenus\Equip\
Tag ProMenu Graphics Location	Defines where the ProMenu GUS graphics associated with tags are stored. (see ATTENTION box following this table)	C:\Displays\ProMenus\Tags\
ProMenu Database File	Defines the storage directory and file name for the ProMenu database file. The ProMenu database file is created and maintained using the ProMenu Configuration utility, described in ProMenu Configuration Utility section.	C:\HWIAC\GUSPro\Pro Menu\ProMenuDatabase.dat
Document Map Database Location	Defines where the Document Map databases are stored. This setting is only required if document maps are used. It is ignored otherwise. The Document Map database files are created and maintained using the Document Map Configuration utility, described in Document Map Configuration Utility section.	C:\HWIAC\GUSPro\DocumentMaps\



ATTENTION

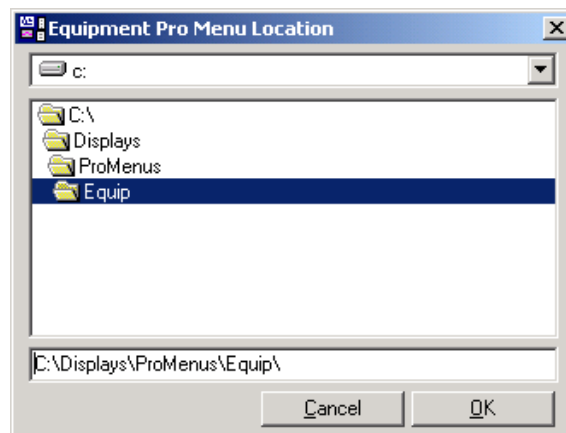
- All directory locations identified in the GUSPro Settings display must be local. The same directory settings must be used in all of the nodes that are running the ProMenu application.
- If the **ProMenu database file** name is changed, you must manually rename the ProMenuDatabase.dat file to suit.
- If a full directory path (like C:\Displays\ProMenus\Tags) is used for a **ProMenu Graphics location**, it overrides the normal GUS search path only for ProMenu invocations. This allows users to maintain the GUSPro graphics as a distinct set of .pct files, independent of the Area configuration.

For sites that prefer to maintain all of their GUS graphics in a directory within the search path, the Equipment ProMenu Graphics Location and Tag ProMenu Graphics Location can be configured as blank; in which case, ProMenu uses the normal GUS search path for invoking displays.

Changing ProMenu File Storage Settings

The directory and file names, initially setup as part of the GUSPro software installation, may be changed either by typing a new entry in the appropriate text box or by clicking the associated **Browse** button. The Location dialog box (Figure 4) may be used to identify the required target drive letter and directory location.

Figure 4 Location Dialog Box



3. ProMenu Configuration

3.2. GUSPro Settings Display

Back in the *GUSPro* Settings display, use the **Reset to Default** button to revert all of the settings to the default entries described in Table 2. (NOTE: The drive letter C:\ is hardcoded in the default directory names.)

Click **OK** to save any changes made to the settings and return to the *GUSPro* Configuration utility. Click **Cancel** to ignore all of the setting changes and return to the *GUSPro* Configuration utility. Click **Apply** to save the setting changes but remain in the *GUSPro* Settings display.

Replication

When the *GUSPro* Settings display is closed, the local *GUSPro.ini* file is created or updated if any changes have been made to the settings. Instructions on how to replicate this file are provided in the Readme file called “*GUSProConfigurationReadme.txt*” located in the %*GUSProDir*% directory. This text file is updated each time the settings in the *GUSPro.ini* file are changed.



TIP

1. The file storage locations and target computers for the *GUSPro* configuration files and databases are dependent on the hardware architecture, which is in place at your site. The replication of these files must fit with your on-site replication strategy.
2. %*GUSProDir*% is a system environment variable containing the name of the parent installation directory for the ProMenu and ProTrend software. If the default install directory was used, the *GUSProDir* directory is C:\Program Files\Honeywell\GUSPro.

4. ProMenu Database Configuration

4.1 Background

ProMenu Rules

ProMenu rules are used to associate tag or equipment names with ProMenu graphics based on characters in the tag or equipment name. For example to specify all tags that contain the string LIC, the ProMenu rule *LIC* could be used. Tags and equipment are separated to allow for a different default ProMenu for tags and for equipment.

ProMenu rules can contain wildcard characters (? = any single character, * = any number of characters). They are used to define which ProMenu is to be called up when an equipment or tag name is passed to the ProMenu application.

For equipment, the master list of Alarm Configuration Manager equipment and equipment groups may be imported from ACM. It is also possible to import a list of equipment names from a Microsoft Excel file. As an aid to ProMenu rule configuration, the ProMenu Configuration utility keeps track of the equipment and equipment group names which have been matched to Equipment ProMenu rules and which have not. This helps you keep track of which rules still need to be defined.

Tag rules are configured in much the same way as equipment rules. Existing tag lists can be imported from a Microsoft Excel file. The ProMenu Configuration utility keeps track of the tags which have been assigned to Tag ProMenu rules and which have not. That way you can ensure that all tags are ultimately assigned to a Tag ProMenu rule.

It is up to the implementer to ensure that every tag, equipment or equipment group name is matched by at least one ProMenu rule.

ProMenu Database

The ProMenu database is where the ProMenu rules are stored. It defines the mapping of ProMenu rules to ProMenu graphics. There must be one instance of the ProMenu database on each station that is using the ProMenus. The database is a flat text file called ProMenuDatabase.dat. By fault, this database is located in C:\HWIAC\GUSPro\Pro Menu. (The database file name and location can be changed if desired.)

ProMenu rules are applied in the order that they appear in the database. As a result, it may be necessary to reorder the ProMenu rules within the database to ensure that the correct ProMenu is invoked for each tag and equipment.

The ProMenu database must only be modified using the ProMenu Configuration utility.

4. ProMenu Database Configuration

4.1. Background

Default ProMenus

A default ProMenu must be defined for each type of ProMenu (that is, tag and equipment). If no default is defined, the default Equipment ProMenu PMEqDefault.pct and the default Tag ProMenu PMTagDefault.pct may be used. A default is indicated in the database by a “*” for the rule.

In the ProMenu database, the default rules are always listed last so they can catch any tag or equipment names that are not matched by any other rule.

Importing Lists from Excel

Equipment, Equipment Groups and Tag Name Lists



ATTENTION

It is recommended that the equipment and equipment group names be imported directly from ACM, rather than from an Excel workbook. This helps to ensure equipment name consistency between ACM and the other applications. Equipment name lists in Excel must only be used if ACM is not available.

As stated earlier, the equipment, equipment group and tag names, that are used to check the ProMenu rules, may be imported from Microsoft Excel files. To do this, the Excel files are set up in a manner similar to that shown in Figure 5.

Figure 5 Sample Equipment List in Excel

The screenshot shows a Microsoft Excel window titled "Microsoft Excel - EquipmentList.xls". The menu bar includes File, Edit, View, Insert, Format, Tools, Data, Window, and Help. The toolbar contains various icons for file operations and formatting. The formula bar shows "A1" and "= NAME". The worksheet grid has columns A through F and rows 1 through 7. The data in the grid is as follows:

	A	B	C	D	E	F
1	NAME					
2	dist002					
3	dist003					
4	dist004					
5	dist005					
6	dist006					
7	exch001					

The status bar at the bottom shows "Ready" and "NUM".

The equipment, equipment group or tag names must be listed in a single column. It does not matter which column the list appears in. The first row in this column (called the column header) must contain a name that identifies the column contents. In Figure 5, the equipment group names are listed under the column header “NAME”. Any column header name may be used, but NAME is the default identified in the ProMenu Configuration utility.



ATTENTION

The column headers must appear in the first row in the worksheet. The load operation fails, if this is not the case.

The equipment, equipment group or tag names may be stored in any worksheet in the workbook. If the default worksheet for the workbook (for example, Sheet1) is used, there is no need to identify it at import time. If a different worksheet is used, it must be identified in the appropriate dialog box before the import can proceed.



TIP

In Excel, the default worksheet is the worksheet that was selected when the workbook file was last saved.

4. ProMenu Database Configuration

4.1. Background

ProMenu Rule Lists

It is also possible to import, predefined ProMenu rules for both Equipment and Tag ProMenus. Figure 6 shows a sample Excel file containing the Tag ProMenu rules for a set of Tag ProMenu graphic files.

Figure 6 Sample Tag ProMenu Rule List in Excel

	A	B	C	D	E	F	G
1	NAME	TYPE	RULE				
2	PMAAnalogIn.pct	TAG	*AI*	*LI*	*fi*	*pi*	*ti*
3	PMAAlarmFI.pct	TAG	*lah*	*pah*			
4							
5							

In this case, the graphic file names and the associated ProMenu type (that is, Equipment or Tag) must each be listed in a single column, with the first row containing the identifying column header name. The default column headers used in the ProMenu Configuration utility, are NAME and TYPE, respectively.

The set of ProMenu rules for each ProMenu graphic must be listed one rule per column in the row containing the associated graphic file name. The first rule must appear in the column with the column header RULE (or some other identifying column header name). When the ProMenu rules are imported, the ProMenu Configuration utility collects every rule in each column until an empty cell is reached.

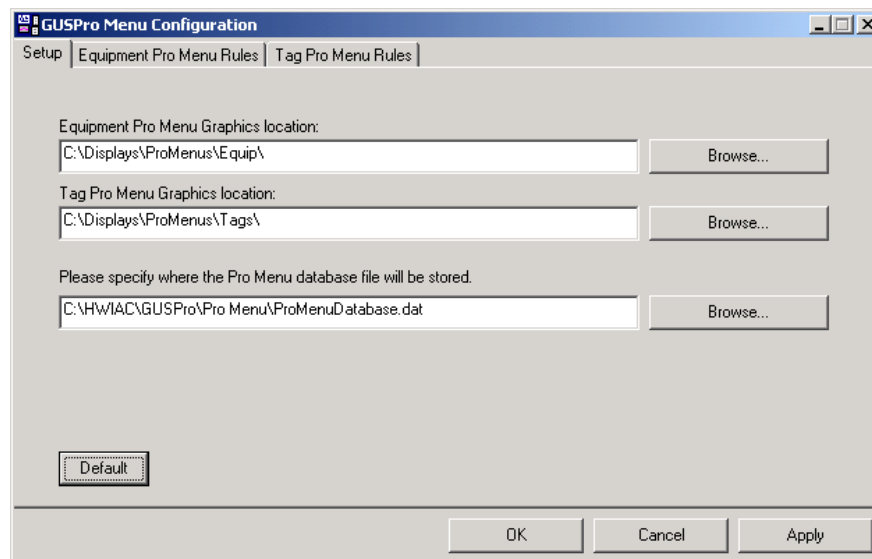
Any worksheet in the workbook may be used as long as it is identified at import time.

4.2 ProMenu Configuration Utility

Purpose

The ProMenu Configuration utility (Figure 7) is used to configure the ProMenu database and to specify the storage locations for the ProMenu files.

Figure 7 ProMenu Configuration Utility



4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Before You Begin

Before running the ProMenu Configuration utility the following items must be defined:

- Pathname catalogue must include the location of the ProMenu graphics. Although this is not used by the Configuration utility, it is necessary for this to be set up before operators can invoke ProMenu displays.
- ProMenus or, at the very least, the names of the ProMenus for both equipment and tags.
- Locations of the ProMenus and the ProMenu databases.
- ACM equipment lists. These are used to define the Equipment ProMenu rules. If the ACM equipment lists are not yet configured, the rules can still be configured but the checks done by the ProMenu Configuration utility does not work.
- Tag database. If the tag database is not built then at least the tag naming convention must be defined. If the tag database is built it can be used (in Microsoft Excel form) as a check to determine if all the tags have been associated appropriately with a ProMenu.

Access

To access the ProMenu Configuration utility, be sure that you are logged on using an account with local administrator privileges. Launch the GUSPro Configuration utility from the Windows **Start** menu by selecting **Programs > Honeywell OPS > GUSPro > GUSPro Configuration**. In the GUSPro Configuration utility, click the **ProMenu Configuration** button.

Functions and Features

The ProMenu Configuration utility is comprised of a series of tabbed displays. When performing the ProMenu configuration tasks, proceed from tab to tab in the order shown. All of the ProMenu configuration tasks are mandatory.

The **OK**, **Cancel** and **Apply** buttons are common to all tabs and provide the following functions:

Button	Function
OK	Writes all changes to the ProMenu database and GUSPro.ini files and closes the Configuration utility.
Cancel	Closes the Configuration utility without applying changes since the last apply.
Apply	Writes all changes to the ProMenu database and GUSPro.ini files without closing the Configuration utility.

ProMenu Rules Tabs

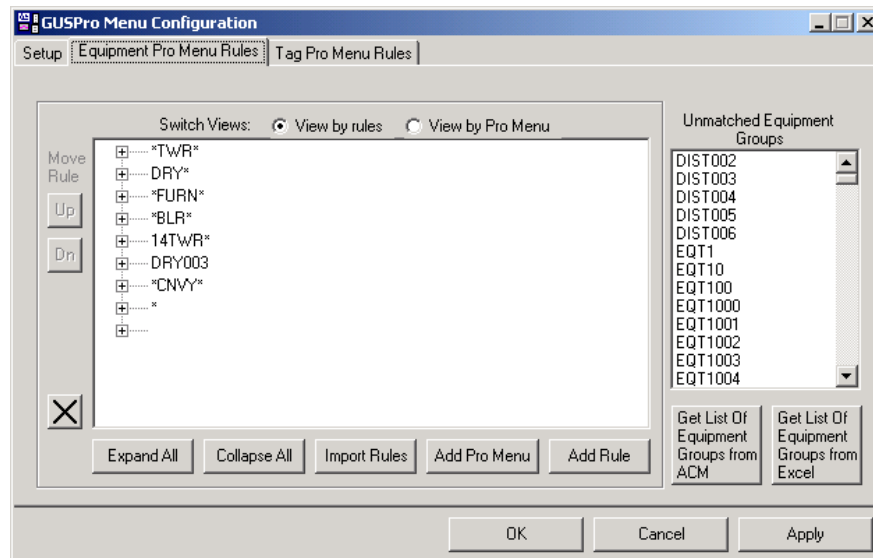
The two ProMenu Rules tabs are used to define ProMenu rules for Equipment and Tag ProMenus. Both tabs have a similar layout and many of the same features.

The ProMenu rules can be viewed either by rules or by ProMenus. When the **View by rules** option is selected, all of the rules currently defined in the ProMenu database file, which are of the appropriate ProMenu type (that is, equipment or tag) are displayed.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Figure 8 Equipment ProMenu Rules Tab - View by Rules

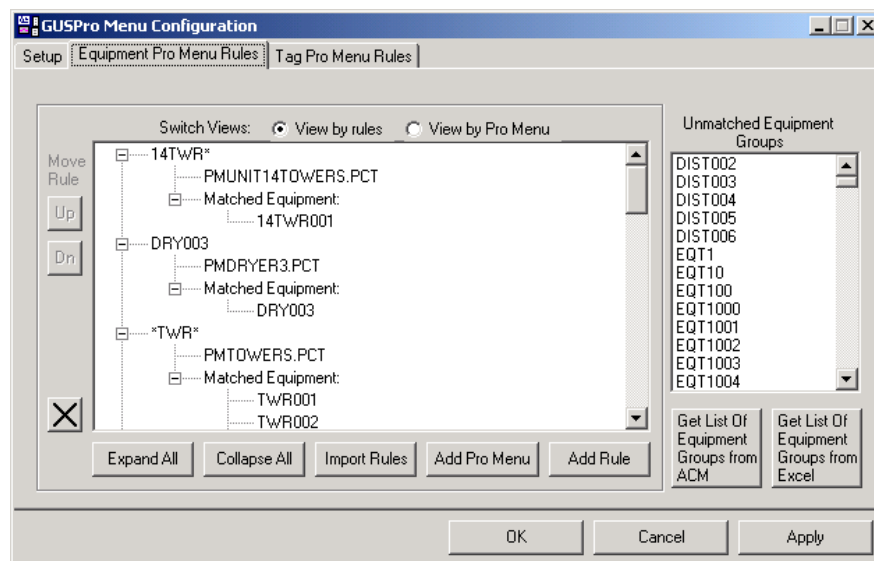


If there are any ProMenu graphic files in the associated ProMenu graphic directory, that have not been assigned to an existing ProMenu rule, blank rules appear at the bottom of the rule list. When expanded, the blank rules show the names of the unassigned ProMenu files. Prior to completing ProMenu configuration, each unassigned ProMenu graphic file must be assigned to a rule. (NOTE: The blank rules are not included in the ProMenu database file. They are provided as an aid to the implementer only.)

Since it is possible for individual pieces of equipment, equipment groups or tags to be matched by more than one rule, and since rules are matched in the order that they appear, it is important for the rules to be organized in an order that makes sense. In the View by rules view the order in which the rules appear in the list can be changed. Generally the rules that are more constraining must appear at the top of the rules list.

For example, in Figure 8 it can be seen that any equipment or equipment group name that matches the 14TWR* rule, would first be matched by the *TWR* rule. Similarly, the DRY* rule would be matched before the DRY003 rule. As a result, the 14TWR* and DRY003 rules, which are more constraining, must be moved up the rules list, as shown in Figure 9.

Figure 9 Rule Order Example

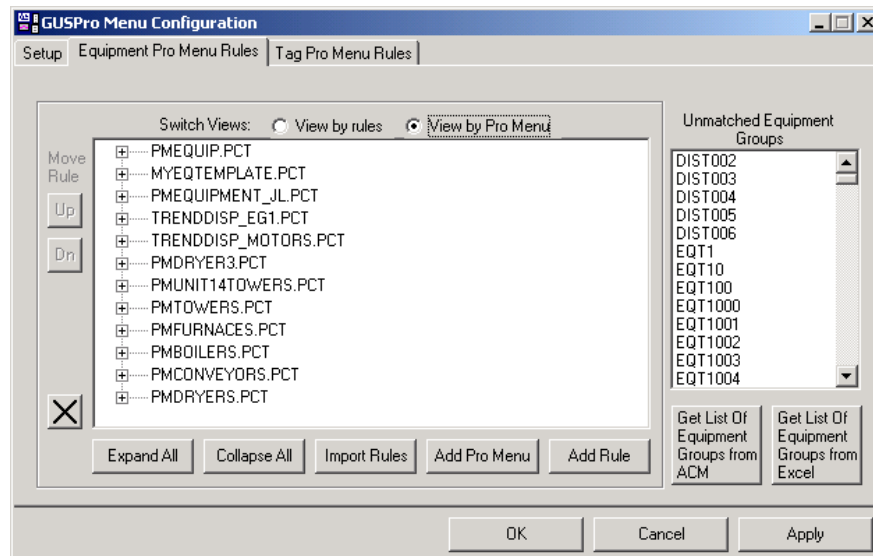


4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

When the **View by ProMenu** option is selected, all ProMenus (in the form of GUS graphic files with a .pct file extension) that exist in the associated ProMenu graphics directory (assigned in the Setup tab) and that have been defined in the ProMenu database file are displayed.

Figure 10 Equipment ProMenu Rules Tab - View by ProMenu



Changes made in the View by ProMenu view are automatically reflected in the View by rules view, and vice versa.

In both views, the following operations can be performed:

- New rules can be added. They can be added singly or imported from an Excel file.
- For Equipment ProMenus, the master list of ACM equipment and equipment groups can be imported from ACM (preferred) or from a Microsoft Excel file. (They are used to check that the set of rules is complete.) Optional.
- For Tag ProMenus, a predefined list of tags can be imported from a Microsoft Excel file. (They are used to check that the set of rules is complete.) Optional.

When the equipment and equipment groups, or tag lists, are imported, any equipment entities or tags that match the pattern for an existing rule, are removed from the associated **“Unmatched”** list and appear beneath the rule. This must help you to remember which rules still need to be defined.

Work Flow

In general, ProMenu configuration must proceed as follows:

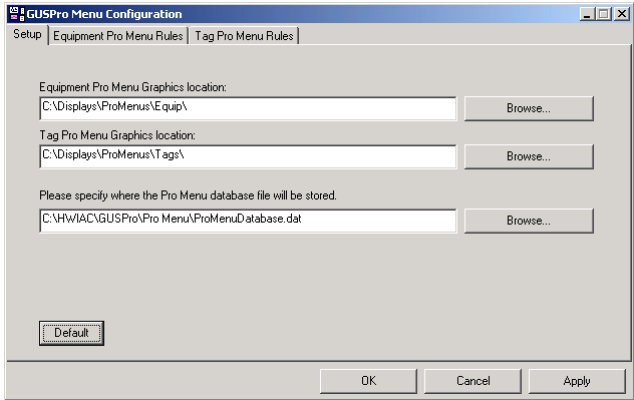
1. Configure the directory locations for the ProMenu database file, Equipment ProMenu graphic files and Tag ProMenu graphic files.
2. Import the predefined equipment and equipment group names either directly from ACM (preferred) or from a Microsoft Excel file, if desired.
3. Create the Equipment ProMenu rules. Rules can either be added one at time or imported from an Excel file.
4. Ensure that every piece of equipment and equipment group in the system is covered by at least one Equipment ProMenu rule and that there are no unmatched equipment and equipment groups.
5. Change the order of the Equipment ProMenu rules to ensure that the equipment and equipment groups are covered by the appropriate rule.
6. If the Equipment ProMenu graphics files have not yet been created, define the list of ProMenus and assign them to Equipment ProMenu rules.
7. Ensure that every Equipment ProMenu graphic is included in at least one Equipment ProMenu rule.
8. Import the predefined tag names from a Microsoft Excel file, if desired.
9. Repeat Steps 3 through 7 to create Tag ProMenu rules for every tag and Tag ProMenu in the system.
10. Save the ProMenu configuration settings in the ProMenu database file and GUSPro.ini file, and replicate the files as directed.

The following sections describe the individual ProMenu configuration tasks in detail.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Define the General ProMenu Settings

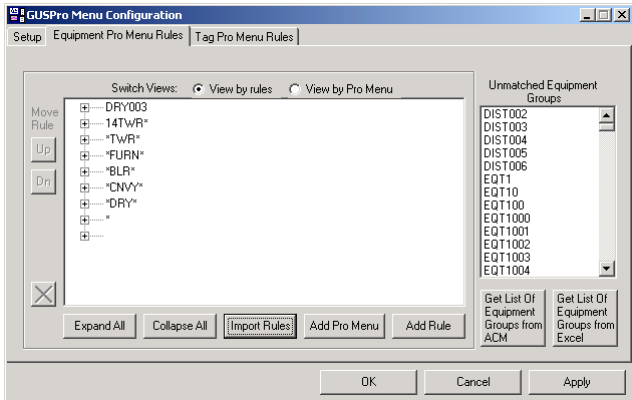
Step	Action
1	In the Setup tab, specify the file storage location for the following ProMenu files: • Equipment ProMenu GUS graphics • Tag ProMenu GUS graphics • ProMenu database file (along with the ProMenuDatabase.dat file name)  The directory names may be changed either by typing a new entry in the appropriate text box or by clicking the associated Browse button. Use the Default button to revert all of the ProMenu settings to the default entries shown in the figure. The same settings can also be configured in the GUSPro Settings display (described in the GUSPro Settings Display section).



ATTENTION

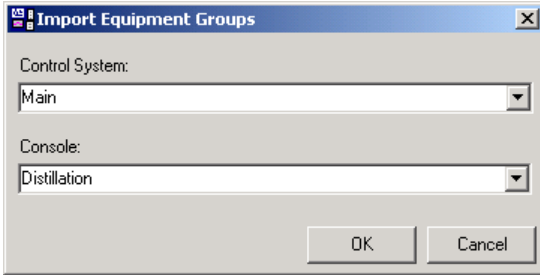
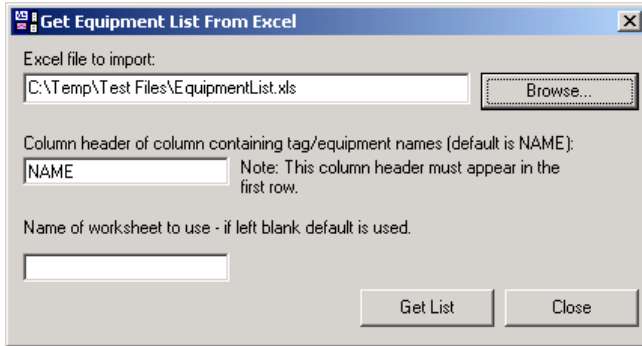
If the ProMenu database file name is changed, a *new* ProMenu database file is created with this file name when the **OK** or **Apply** buttons are clicked. It is up to the implementer to delete all copies of the ProMenu database file with the old file name that might have been created in previous ProMenu configuration sessions.

Configure the Equipment ProMenu Rules

Step	Action
1	Open the Equipment ProMenu Rules tab. By default, the View by rules view is displayed which shows all of the Equipment ProMenu rules currently defined in the ProMenu database file.  To see the details for each rule, double-click the rule name in the list or use the Expand All and Collapse All buttons. When a rule is expanded, the ProMenu graphic file (.pct files) that is assigned to the rule is displayed. Once a list of equipment or equipment group names has been loaded, those names that match the existing rule patterns are also displayed with the matching rule.
2 a)	As an aid to configuring the Equipment ProMenu Rules, load the master list of equipment and equipment groups from ACM by clicking Get List of Equipment Groups from ACM. (This is the preferred method for importing the equipment and equipment groups list and it can be performed from the View either by rules or View by ProMenu view.) (Optional.)

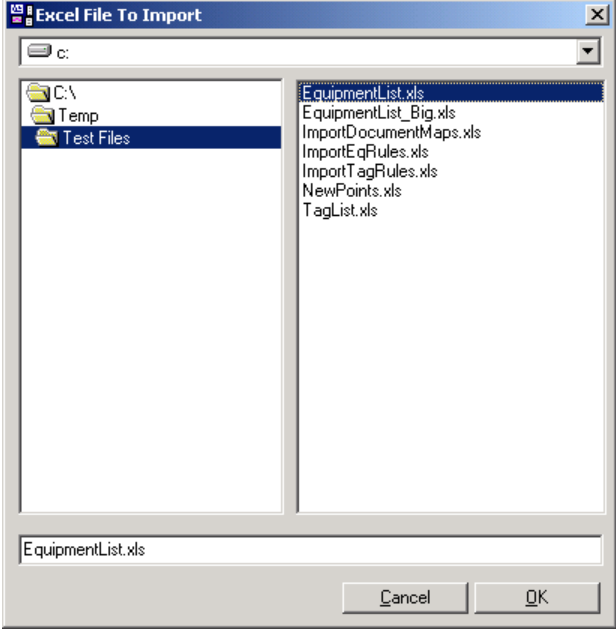
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
2 b)	In the Import Equipment Groups dialog box, identify the ACM control system and ACM console, and click OK.  The equipment and equipment group names are then collected directly from the specified ACM console. (NOTE: If the ACM Alarm Manager Server is not available, the Get List of Equipment Groups from ACM operation fails.)
2 c)	If ACM is unavailable or has not yet been configured, the list of equipment and equipment groups can also be collected from a Microsoft Excel file by clicking Get List of Equipment Groups from Excel. This invokes the Get Equipment List From Excel dialog box. (This operation can be performed from either view.)  In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to launch the Excel File To Import dialog box. This dialog box may be used to locate the required Excel file.

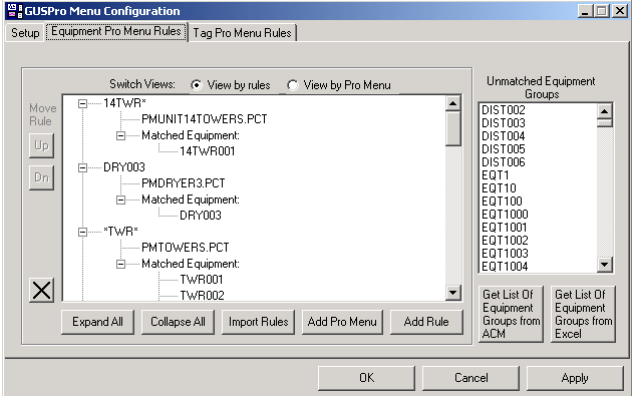
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
	 Back in the Get Equipment List From Excel dialog box, identify the column in the Excel file that contains the equipment or equipment group names, by typing the column-heading name in the Column header text box. If the equipment and equipment group names are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. Click Get List to load the equipment and equipment groups identified in the Excel workbook or click Close to cancel the load operation.

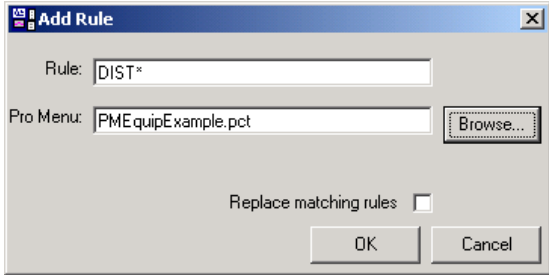
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
2 d)	When a list of equipment and equipment group names are loaded from either ACM or an Excel file, the ProMenu Configuration utility scans the list of equipment and equipment groups names searching for those names, which match the patterns specified by the existing rules. When a match is found, the equipment or equipment group name is added to the Matched Equipment list for the rule. Any equipment and equipment groups that do not match an existing Equipment ProMenu rule, are displayed in the Unmatched Equipment Groups list. If no rules have been defined, all of the equipment and equipment groups appear in the Unmatched Equipment Groups list.  NOTE: Each time a list of equipment names is imported, any existing equipment names in the Equipment ProMenu Rules tab are replaced. The new equipment names are not appended to an existing list.

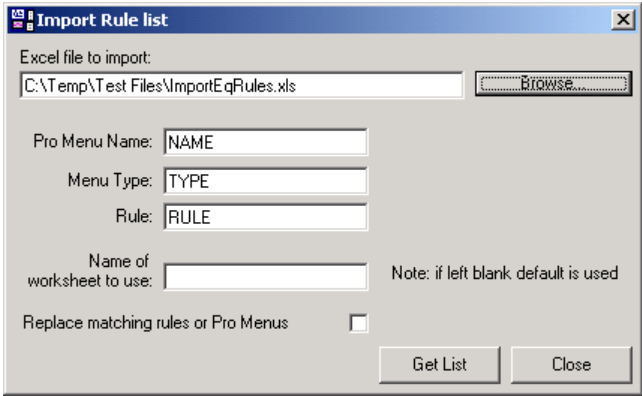
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
3 a)	To add a single new rule, click Add Rule to invoke the Add Rule dialog box. (This operation can be performed from either view.)  In the Rule text box, type the rule name. Rules can contain wildcard card characters (? = any single character, * = any number of characters). In the ProMenu text box, either type the associated ProMenu graphic name (including the .pct file extension) or click Browse to select the required ProMenu graphic file name in the associated ProMenu graphics directory. To have the new rule, replace a previously defined rule with the same name, click the Replacing matching rules check box to select it. Click OK to accept the new rule and return to the ProMenu Rules tab. The new rule is automatically added to the bottom of the rules list, directly above the default ('*') rule. Click Cancel to return to the ProMenu Rules tab without saving the new rule.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
3 b)	To import a predefined set of rules from an Excel file, click Import Rules. This invokes the Import Rule list dialog box. (This operation can be performed from either view.)  In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to locate the required Excel file. Identify the columns in the Excel file that contain the ProMenu graphic file names, ProMenu type (that is, Equipment) and the rule names, by typing the associated column-heading name in the appropriate text box. If the rules are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. To have the new set of rules replace any previously defined rules with the same names, click the Replacing matching rules or ProMenus check box to select it. Click Get List to import the rules identified in the Excel workbook or click Close to cancel the import operation. The new rules are automatically added to the bottom of the rules list, directly above the default ("*") rule.
3 c)	When a rule is created or a set of rules is imported, the ProMenu Configuration utility scans the current list of equipment and equipment group names searching for those names, which match the patterns specified by the new rules. When a match is found, the equipment or equipment group name is removed from the Unmatched Equipment Groups list and is added to the Matched Equipment list for the matching rule.

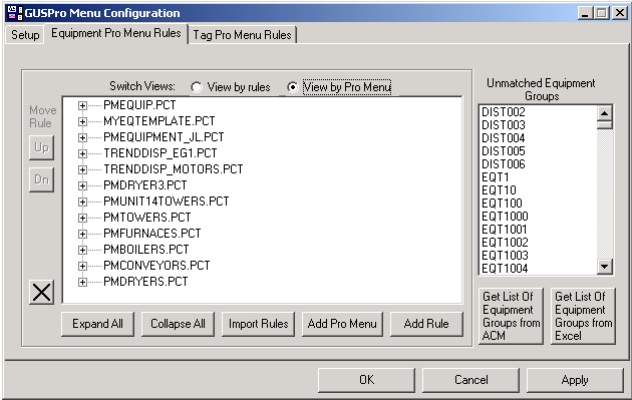
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
4	To remove an existing rule, select the rule from the list and click the X button. Any equipment or equipment group names that matched the deleted rule pattern are either returned to the Unmatched Equipment Groups list or are added to another existing rule. This second operation would only occur if another rule pattern match is found further down the list of rules.
5	To modify an existing rule, select the rule name, pause and then single-left-click the rule name. When the edit box is displayed, make the desired changes and press ENTER. The ProMenu Configuration utility rescans the current list of equipment and equipment group names. The following results are possible: - Any equipment or equipment group names that no longer match the modified rule pattern are removed from the Matched Equipment list for the rule. These equipment or equipment group names are either returned to the Unmatched Equipment Groups list or are added to another existing rule. - Any equipment or equipment group names in the Unmatched Equipment Groups list that now match the new rule pattern are added to the Matched Equipment list for the rule.
6	To change the order in which the rules are listed, ensure that the View by rules option is selected. (This operation cannot be performed from the View by ProMenu view.) Select the rule that needs to be reordered and use the Up and Dn Move Rule buttons to move it. Each time the rule order is changed, the ProMenu Configuration utility rescans the list of equipment and equipment group names. This may cause some of equipment and equipment group names to change the rule that they are matched with.
7	Ensure that every equipment and equipment group in the system is covered by at least one Equipment ProMenu rule and that there are no unmatched equipment and equipment groups.

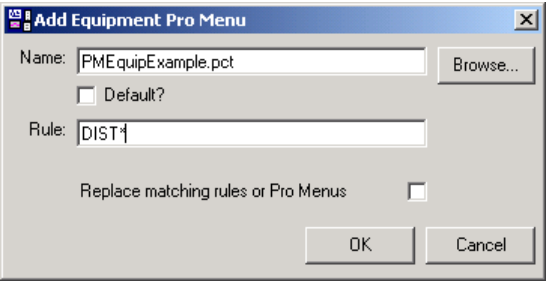
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
8	Select the View by ProMenu option to show current set of Equipment ProMenu graphic files that exist in the Equipment ProMenu graphics directory or that have been defined in the ProMenu database file.  To see the details for each ProMenu, double-click the ProMenu name in the list or use the Expand All and Collapse All buttons. When a ProMenu is expanded, the rule to which the ProMenu has been assigned is displayed, along with the equipment or equipment group names that match the associated rule pattern. If a ProMenu has not yet been assigned to a rule, there are no details (that is, no rules or matching equipment names) available for the ProMenu.

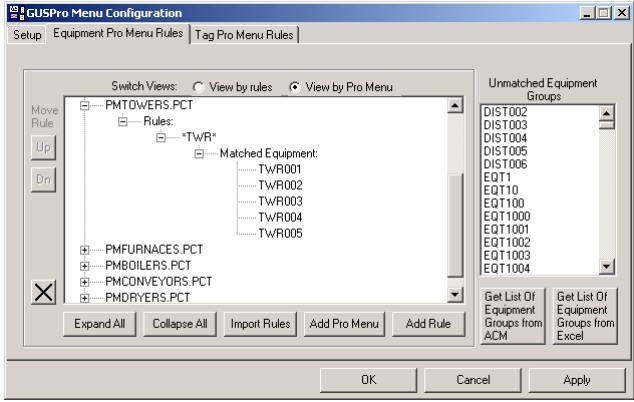
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

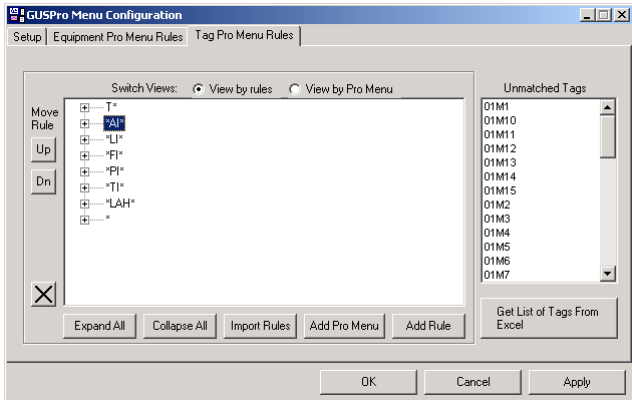
Step	Action
9 a)	To add a ProMenu to a rule, click Add ProMenu to invoke the Add Equipment ProMenu dialog box. (This operation can be performed from either view.)  In the Name text box, either type the ProMenu graphic name (including the .pct file extension) or click Browse to select the required ProMenu graphic file name in the associated ProMenu graphics directory. Select the Default? option, if the ProMenu graphic being assigned, is to act as the default ProMenu for all of the Equipment ProMenus. When this happens the Rule text box automatically displays the * character and cannot be changed. If this is not the default rule, type the rule name in the Rule text box. (New rules may be added here.) To have the new ProMenu or rule replace a previously defined ProMenu or rule with the same name, click the Replacing matching rules or ProMenus check box to select it. Click OK to accept the new ProMenu and return to the ProMenu Rules tab, or Cancel to return to the ProMenu Rules tab without saving the new ProMenu.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

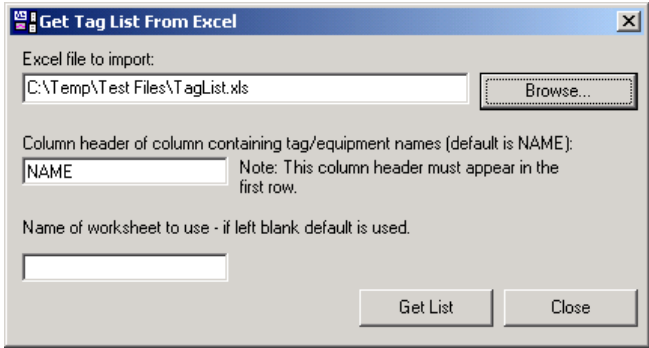
Step	Action
9 b)	If new rules are created as part of the add ProMenu operation, the ProMenu Configuration utility rescans the list of equipment and equipment group names looking for matches. 
10	To modify a ProMenu name after it has been added, select the ProMenu name, pause and then single-left-click the text. When the edit box is displayed, make the desired changes and press ENTER. To delete an existing ProMenu, select the ProMenu from the list and click the X button. (Both of these operations can be performed from either view.)
11	Ensure that every Equipment ProMenu graphic is included in at least one Equipment ProMenu rule. In the View by rules view, ensure that there are no ProMenu graphic files listed in the blank rules at the bottom of the hierarchy. NOTE: Unassigned ProMenu graphic files, that remain in the blank rules after configuration is complete, are not added to the ProMenu database file.

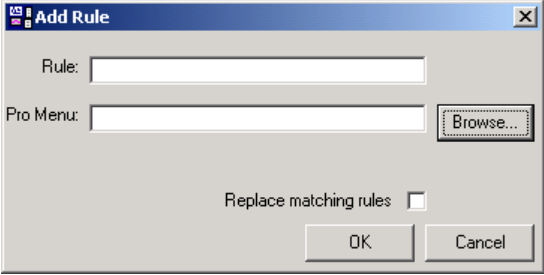
Configure the Tag ProMenu Rules

Step	Action
1	Open the Tag ProMenu Rules tab. By default, the View by rules view is displayed which shows all of the Tag ProMenu rules currently defined in the ProMenu database file.  To see the details for each rule, double-click the rule name in the list or use the Expand All and Collapse All buttons. When a rule is expanded, the ProMenu graphic file that is assigned to the rule is displayed. Once a list of tags has been loaded, those names that match the existing rule patterns are also displayed with the matching rule.

4. ProMenu Database Configuration

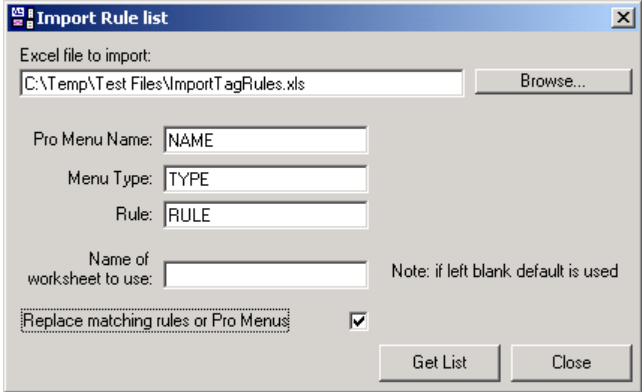
4.2. ProMenu Configuration Utility

Step	Action
2 a)	As an aid to configuring the Tag ProMenu Rules, load the list of tags from a Microsoft Excel file by clicking Get List of Tags from Excel. This invokes the Get Tag List From Excel dialog box. (This operation can be performed from either view.) (Optional.)  In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to launch the Excel File To Import dialog box. This dialog box may be used locate the required Excel file. Back in the Get Tag List From Excel dialog box, identify the column in the Excel file that contains the tag names, by typing the column-heading name in the Column header text box. If the tag names are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. Click Get List to load the tags identified in the Excel workbook or click Close to cancel the load operation.
2 b)	When a list of tags is loaded from an Excel file, the ProMenu Configuration utility scans the list of tag names searching for those names, which match the patterns specified by the existing rules. When a match is found, the tag name is added to the Matched Tags list for the rule. Any tag names that do not match an existing Tag ProMenu rule, are displayed in the Unmatched Tags list. If no rules have been defined, all of the tags appear in the Unmatched Tags list. NOTE: Each time a list of tags is imported, any existing tag names in the Tag ProMenu Rules tab are replaced. The new tag names are not appended to an existing list.

Step	Action
3 a)	To add a single new rule, click Add Rule to invoke the Add Rule dialog box. (This operation can be performed from either view.)  In the Rule text box, type the rule name. In the ProMenu text box, either type the associated ProMenu graphic name (including the .pct file extension) or click Browse to select the required ProMenu graphic file name in the associated ProMenu graphics directory. To have the new rule, replace a previously defined rule with the same name, click the Replacing matching rules check box to select it. Click OK to accept the new rule and return to the ProMenu Rules tab. The new rule is automatically added to the bottom of the rules list, directly above the default ('*') rule. Click Cancel to return to the ProMenu Rules tab without saving the new rule.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
3 b)	To import a predefined set of rules from an Excel file, click Import Rules. This invokes the Import Rule list dialog box. (This operation can be performed from either view.)  In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to locate the required Excel file. Identify the columns in the Excel file that contain the ProMenu graphic file names, ProMenu type (that is, Tag) and the rule names, by typing the associated column-heading name in the appropriate text box. If the rules are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. To have the new set of rules replace any previously defined rules with the same names, click the Replacing matching rules or ProMenus check box to select it. Click Get List to import the rules identified in the Excel workbook or click Close to cancel the import operation. The new rules are automatically added to the bottom of the rules list, directly above the default ("*") rule.
3 c)	When a rule is created or a set of rules is imported, the ProMenu Configuration utility scans the current list of tag names searching for those names, which match the patterns specified by the new rules. When a match is found, the tag name is removed from the Unmatched Tags list and is added to the Matched Tags list for the matching rule.

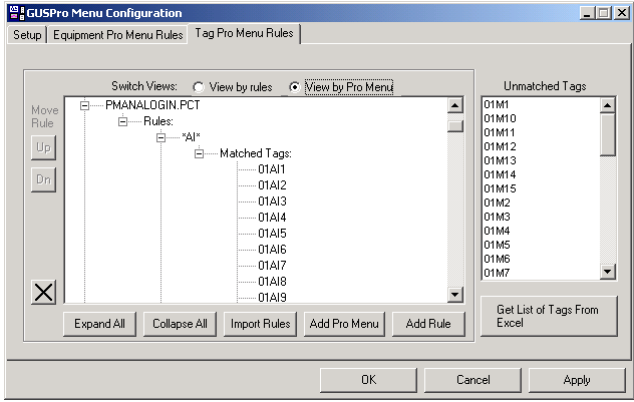
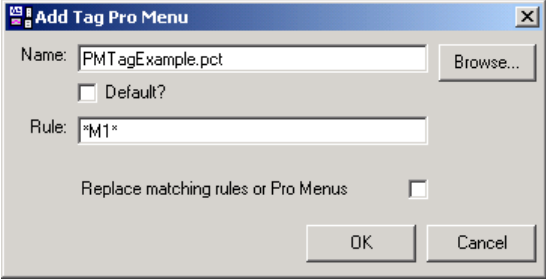
4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
4	To remove an existing rule, select the rule from the list and click the X button. Any tag names that matched the deleted rule pattern are either returned to the Unmatched Tags list or are added to another existing rule.
5	To modify an existing rule, select the rule name, pause and then single-left-click the rule name. When the edit box is displayed, make the desired changes and press ENTER. The ProMenu Configuration utility rescans the current list of tag names. The following results are possible: - Any tag names that no longer match the modified rule pattern are removed from the Matched Tags list for the rule. These tag names are either returned to the Unmatched Tags list or are added to another existing rule. - Any tag names in the Unmatched Tags list that now match the new rule pattern are added to the Matched Tags list for the rule.
6	To change the order in which the rules are listed, ensure that the View by rules option is selected. (This operation cannot be performed from the View by ProMenu view.) Select the rule that needs to reordered and use the Up and Dn Move Rule buttons to move it. Each time the rule order is changed, the ProMenu Configuration utility rescans the list of tag names. This may cause some of tag names to change the rule that they are matched with.
7	Ensure that every tag in the system is covered by at least one Tag ProMenu rule and that there are no unmatched tags.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
8	Select the View by ProMenu option to show current set of Tag ProMenu graphic files that exist in the Tag ProMenu graphics directory and that have been defined in the ProMenu database file.  To see the details for each ProMenu, double-click the ProMenu name in the list or use the Expand All and Collapse All buttons. When a ProMenu is expanded, the rule to which the ProMenu has been assigned is displayed, along with the tag names that match the associated rule pattern. If a ProMenu has not yet been assigned to a rule, there are no details (that is, no rules or matching equipment names) available for the ProMenu.
9 a)	To add a ProMenu to a rule, click Add ProMenu to invoke the Add Tag ProMenu dialog box. (This operation can be performed from either view.)  In the Name text box, either type the ProMenu graphic name (including the .pct file extension) or click Browse to select the required ProMenu graphic file name in the associated ProMenu graphics directory.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Step	Action
	Select the Default? option, if the ProMenu graphic being assigned, is to act as the default ProMenu for all of the Tag ProMenus. When this happens the Rule text box automatically displays the * character and cannot be changed. If this is not the default rule, type the rule name in the Rule text box. (New rules may be added here.) To have the new ProMenu rule replace a previously defined rule with the same name, click the Replacing matching rules or ProMenus check box to select it. Click OK to accept the new ProMenu and return to the ProMenu Rules tab, or Cancel to return to the ProMenu Rules tab without saving the new ProMenu.
9 b)	If new rules are created as part of the add ProMenu operation, the ProMenu Configuration utility rescans the list of tag names looking for matches.
10	To modify a ProMenu name after it has been added, select the ProMenu name, pause and then single-left-click the text. When the edit box is displayed, make the desired changes and press ENTER. To delete an existing ProMenu, select the ProMenu from the list and click the X button. (Both of these operations can be performed from either view.)
11	Ensure that every Tag ProMenu graphic is included in at least one Tag ProMenu rule. In the View by rules view, ensure that there are no ProMenu graphic files listed in the blank rules at the bottom of the hierarchy. NOTE: Unassigned ProMenu graphic files that remain in the blank rules after configuration is complete are not added to the ProMenu database file.

4. ProMenu Database Configuration

4.2. ProMenu Configuration Utility

Replicate the ProMenu Configuration Files

Once all of the ProMenu Configuration utility tabs have been filled in, the ProMenu database file (ProMenuDatabase.dat) and the updated GUSPro.ini file, must be manually replicated to the appropriate computers in the system.

Step	Action
1	In the ProMenu Configuration utility, click OK to close the utility and return to the GUSPro Configuration utility display. This causes a ProMenu readme file (GUSProMenuConfigurationReadme.txt) to be created (or updated) in the %GUSProDir% directory. If any of the ProMenu file storage or database name settings were changed in the Setup tab, closing the ProMenu Configuration utility also causes the GUSPro.ini file to be updated. (The location of the GUSPro.ini file is determined by the %GUSProDir% environment variable.)
2	Review the contents of the GUSProMenuConfigurationReadme.txt file and carry out the instructions provided as to where the ProMenuDatabase.dat and GUSPro.ini files need to be copied.



ATTENTION

It is important that the file locations, specified during ProMenu configuration, be the same on the nodes that are accessing the ProMenu database file and running the ProMenu graphics.



TIP

%GUSProDir% is a system environment variable, which contains the name of the parent installation directory for the ProMenu and ProTrend software. If the default install directory was used, the GUSProDir directory is C:\Program Files\Honeywell\GUSPro.

5. Document Map Configuration

5.1 Background

Document Maps

When documents, that do not follow the ProMenu document naming convention, are to be called from ProMenu graphics, Document Maps can be built that specify which document is opened for a given tag, equipment or equipment group name. Much like ProMenu rules, Document Map rules are used to match tag, equipment or equipment group names to documents.



TIP

If all of the documents that are going to be accessed from ProMenus follow the ProMenu naming convention:

```
<PrefixString>_<Name>.<DocumentExtension>
```

or if no documents are going to be accessed from the ProMenus, then Document Mapping is not necessary. In either of these cases, Document Map configuration can be skipped.

Document Map Rules

Document Map rules are used to associate tag or equipment names with the individual documents of each type, based on characters in the tag or equipment name. As with ProMenu rules, the Document Map rules can contain wildcard characters.

Existing equipment and tag name lists can be imported into the Document Map Configuration utility as an aid to ensuring that every tag, equipment or equipment group name is matched by at least one Document Map rule.

Document Map Databases

The Document Map databases define the mapping of Document Map rules to documents. One Document Map database must exist for every document type being mapped. For example, there may be a startup procedures Document Map and a shutdown procedures Document Map.

5. Document Map Configuration

5.1. Background

The Document Map databases are flat text files, which use the following naming convention:

`<Type>_DocumentMap.dat`

where Type is the type of Document Map. For example a Document Map built for P&IDs might be called “PID_DocumentMap.dat”.

Document Map rules are applied in the order that they appear in the database. As a result, it may be necessary to reorder the Document Map rules within each Document Map database to ensure that the correct document is invoked for each tag and equipment.

The Document Map databases must only be modified using the Document Map Configuration utility.

Importing Lists from Excel

Equipment, Equipment Groups and Tag Name Lists

In the same way as the ProMenu configuration, the equipment, equipment group and tag names, that are used to check the Document Map rules, may be imported from Microsoft Excel files. The exact same Excel files (described in the “Importing Lists from Excel” section) may be used for Document Map configuration.

Document Map Lists

It also possible to import, predefined Document Maps for multiple document types. Figure 11 shows a sample Excel file containing a Document Map list.

Figure 11 Sample Document Map List in Excel

	A	B	C	D	E	F
1	TYPE	DOCUMENT	RULE			
2	PID	C:\Documents\RP-986753.doc	*145*	*146*	*250*	*251*
3	PID	C:\Documents\RP-986754.doc	*EX*	*TWR*		
4	PID	C:\Documents\RP-986755.doc	*tur*	*dry*		
5	PID	C:\Documents\RP-986756.doc	*furn*	*tic*		
6	PID	C:\Documents\RP-986757.doc	*cnvy*	*pic*		

In this case, the document type and the associated documents must each be listed in a single column, with the first row containing the identifying column header name. The default column headers used in the Document Map Configuration utility, are TYPE and DOCUMENT, respectively.

The set of Document Map rules for each document, must be listed one rule per column in the row containing the associated document name. The first rule must appear in the column with the column header RULE (or some other identifying column header name). When the Document Maps are imported, the Document Map Configuration utility collects every rule in each row until an empty cell is reached.

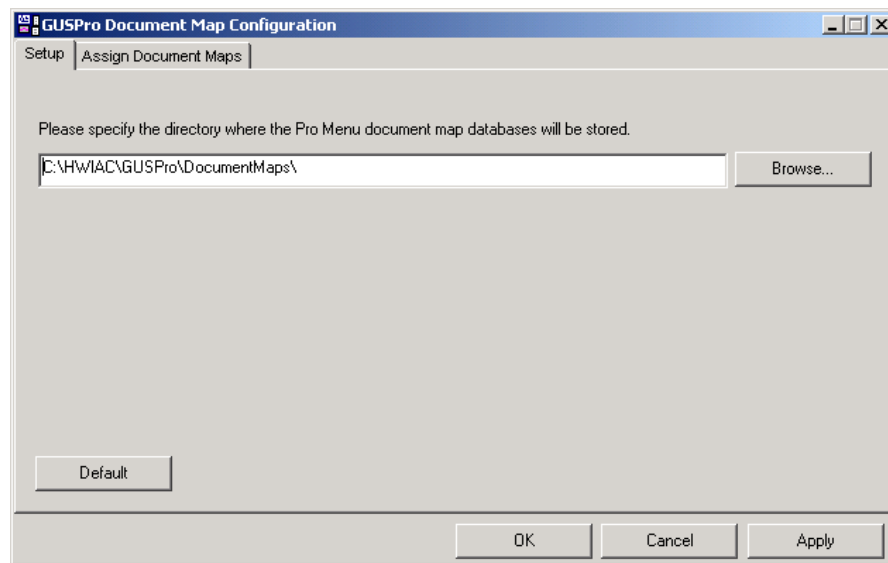
Any worksheet in the workbook may be used as long as it is identified at import time.

5.2 Document Map Configuration Utility

Purpose

The Document Map Configuration utility (Figure 12) is used to create the Document Map databases and to specify the storage location for the Document Map database files.

Figure 12 Document Map Configuration Utility



Before You Begin

Before running the Document Map Configuration utility the following items must be defined:

- Location where the Document Map databases are to be stored.
- Documents, or at least the names of the documents, for which the Document Maps need to be created, grouped by document type.
- ACM equipment lists. If the ACM equipment lists are not yet configured, the Document Maps can still be configured but the checks done by the Document Map Configuration utility does not work.

- Tag database. If the tag database is not built then at least the tag naming convention must be defined. If the tag database is built it can be used (in Microsoft Excel form) as a check to determine if all the tags have been associated appropriately with a Document Map.

Access

To access the Document Map Configuration utility, be sure that you are logged on using an account with local administrator privileges. Launch the GUSPro Configuration utility from the Windows **Start** menu by selecting **Programs > Honeywell OPS > GUSPro > GUSPro Configuration**. In the GUSPro Configuration utility, click the **Document Map Configuration** button.

Functions and Features

The Document Map Configuration utility is comprised of two tabbed displays. When performing the Document Map configuration tasks, proceed from tab to tab in the order shown. All of the Document Map configuration tasks are mandatory.

The **OK**, **Cancel** and **Apply** buttons are common to all tabs and provide the following functions:

Button	Function
OK	Creates or updates the Document Map database files, updates the GUSPro.ini file and closes the Configuration utility.
Cancel	Closes the Configuration utility without applying changes since the last apply.
Apply	Creates or updates the Document Map database files and updates the GUSPro.ini file without closing the Configuration utility.

Assign Document Maps Tab

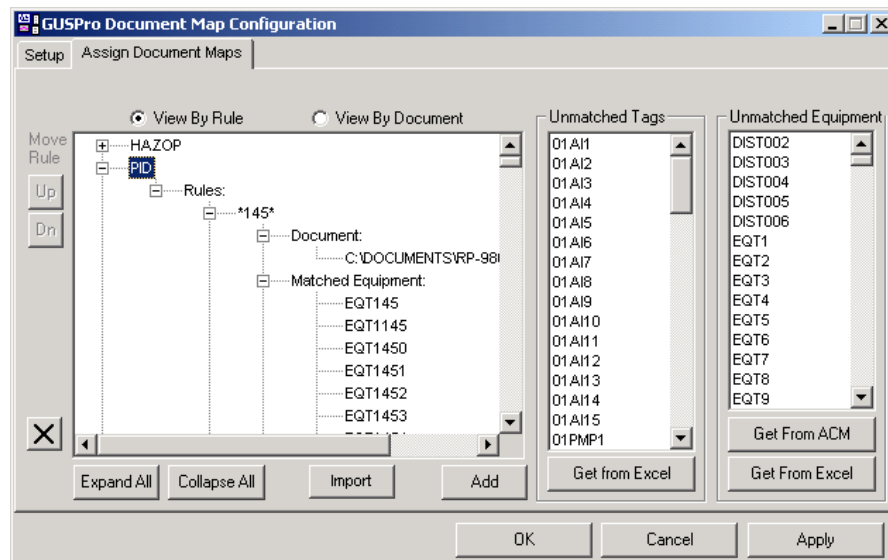
The Assign Document Maps tab is used to define the Document Maps. The tab layout and features are similar to those provided by the ProMenu Configuration utility.

The Document Maps can either be viewed by rules or by documents. When the **View by Rule** option is selected, all of the Document Map rules currently defined in each of the existing Document Map database files are displayed.

5. Document Map Configuration

5.2. Document Map Configuration Utility

Figure 13 Assign Document Maps Tab - View by Rule



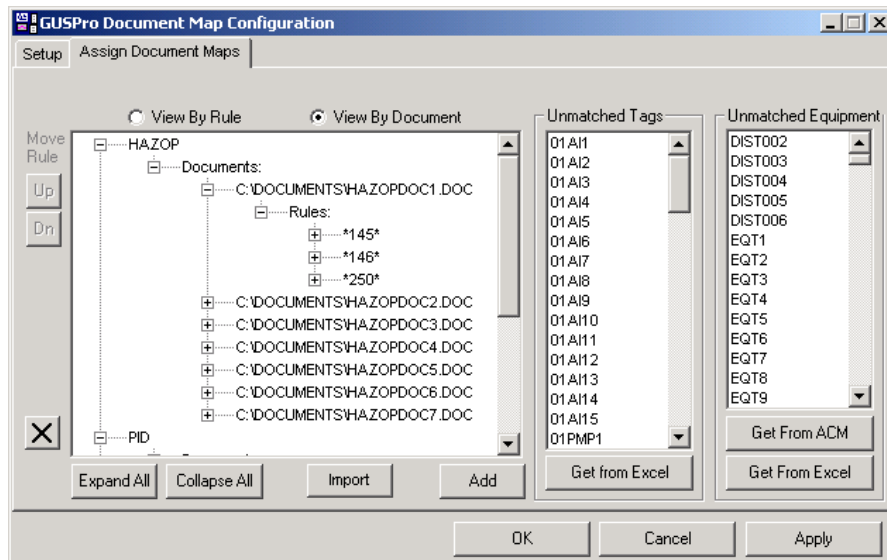
In this view, each type of document is displayed with its own hierarchy. (Each document type hierarchy represents a separate Document Map database.) Beneath the document type name, the Document Map rules defined for the documents of that document type are displayed in their configured order. Each rule is then displayed with the following information:

- The document to which the rule is assigned.
- The equipment and equipment group names that match the rule pattern. (Only displayed if the equipment and equipment group names have been loaded.)
- The tag names that match the rule pattern. (Only displayed if the tags have been loaded.)

As with the ProMenu rules, it is possible for individual pieces of equipment, equipment groups or tags to be matched by two or more Document Map rules. As a result, it is important for the rules in each document type to be organized in an order that makes sense. In the View by Rule view the order in which the rules appear for each document type can be changed. Generally the rules that are more constraining must be moved up in the list of rules for a given document type.

When the **View by Document** option is selected, the set of documents assigned to each document type that are currently defined in the Document Map databases, are displayed.

Figure 14 Assign Document Maps Tab - View by Document



In this view, each type of document is once again displayed with its own hierarchy. Beneath the document type name, the documents assigned to that document type are displayed. Each document is then displayed with the following information:

- The rules assigned to the document.
- The equipment and equipment group names that match the set of rule patterns. (Only displayed if the equipment and equipment group names have been loaded.)
- The tag names that match the set of rule patterns. (Only displayed if the tags have been loaded.)

Changes made in the View by Rule view are automatically reflected in the View by Document view, and vice versa.

5. Document Map Configuration

5.2. Document Map Configuration Utility

In both views, the following operations can be performed:

- New document types can be added or imported from a predefined list. (This results in a new Document Map database being created for each new document type.)
- New documents can be added to a document type.
- New rules can be added for a document.
- The master list of ACM equipment and equipment group names can be imported from ACM (preferred) or from a Microsoft Excel file. (They are used to check that the set of rules is complete.) Optional.
- A predefined list of tags can be imported from a Microsoft Excel file. (They are used to check that the set of rules is complete.) Optional.

When the equipment and equipment groups, or tag lists, are imported, any equipment entities or tags that match the pattern for an existing rule, are removed from the associated “**Unmatched**” list and appear beneath the rule. This must help you to remember which rules still need to be defined.



TIP

Within a given document type, each Document Map rule can only be assigned to *one* document. However, the same Document Map rule can be used for multiple document types.

Work Flow

In general, Document Map configuration must proceed as follows:

1. Configure the directory location for the Document Map database files.
2. Import the predefined tag names from a Microsoft Excel file, if desired.
3. Import the predefined equipment and equipment group names either directly from ACM (preferred) or from a Microsoft Excel file, if desired.
4. Define the types of documents for which Document Maps are required. Document types can either be added one at time or imported from an Excel file.
5. Add the set of documents for each document type.
6. Create the Document Map rules for each document.
7. For each document type, ensure that the appropriate equipment and equipment groups are covered by the required Document Map rules.

8. For each document type, ensure that the appropriate tags are covered by the required Document Map rules.
9. For each document type, change the order of the Document Map rules to ensure that the equipment, equipment groups and tags are covered by the correct rule.
10. Save the Document Map configuration settings in the Document Map database files and GUSPro.ini file, and replicate the files as directed.

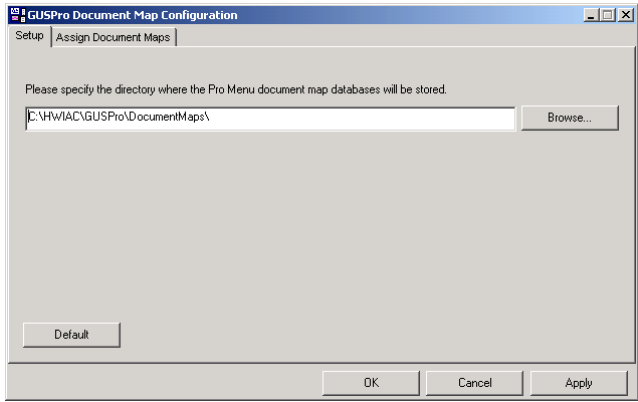


TIP

The individual Document Map rules can apply to both equipment and tags.

The following sections describe the individual Document Map configuration tasks in detail.

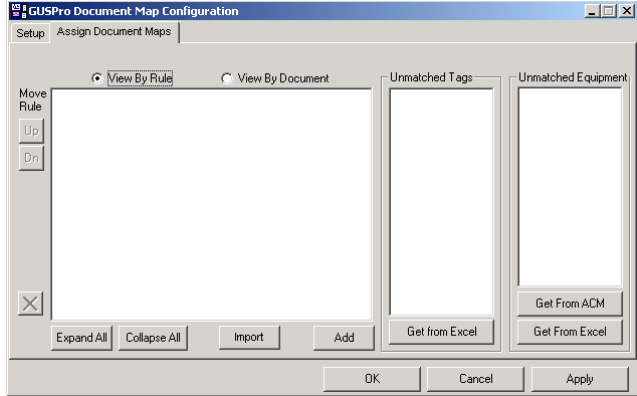
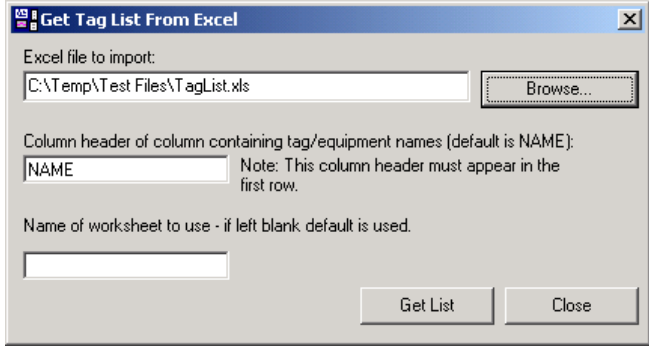
Define the General Document Map Settings

Step	Action
1	In the Setup tab, enter the file storage location for the Document Map database files.  The directory may be changed either by typing a new entry in the text box or by clicking the Browse button. The same directory setting can also be configured in the GUSPro Settings display (described in the GUSPro Settings Display section). Use the Default button to revert to the default Document Map database directory shown in the figure.

5. Document Map Configuration

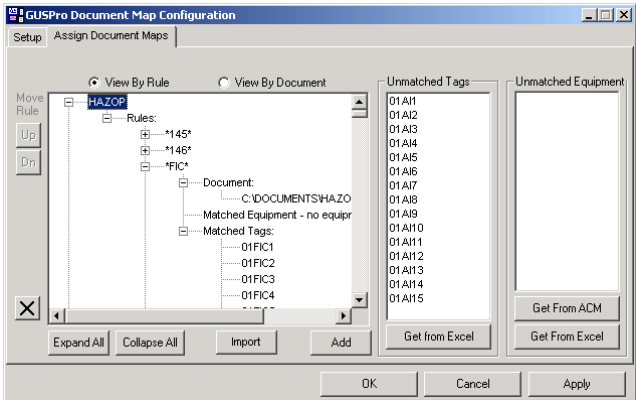
5.2. Document Map Configuration Utility

Configure the Document Maps

Step	Action
1	Open the Assign Document Maps tab. By default, the View by Rule view is displayed.  To see the details for each document type and rule, double-click the name in the list or use the Expand All and Collapse All buttons.
2 a)	As an aid to configuring the Document Map rules, load the list of tags from a Microsoft Excel file by clicking the Get from Excel button in the Unmatched Tags display region. This invokes the Get Tag List From Excel dialog box. (This operation can be performed from either the View by Rule or View by Document view.) (Optional.)  In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to launch the Excel File To Import dialog box. This dialog box may be used locate the required Excel file.

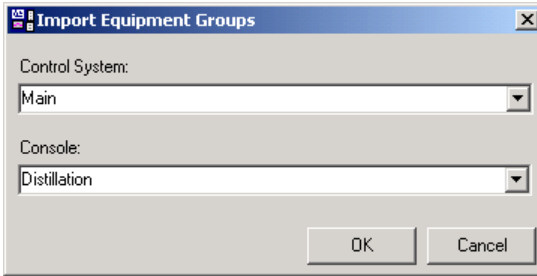
5. Document Map Configuration

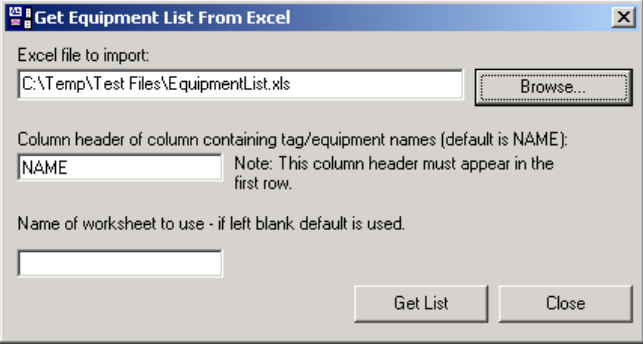
5.2. Document Map Configuration Utility

Step	Action
	Back in the Get Tag List From Excel dialog box, identify the column in the Excel file that contains the tag names, by typing the column-heading name in the Column header text box. If the tag names are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. Click Get List to load the tags identified in the Excel workbook or click Close to cancel the load operation.
2 b)	When a list of tags is loaded from an Excel file, the Document Map Configuration utility scans the list of tag names searching for those names, which match the patterns specified by the existing rules. When a match is found, the tag name is added to the Matched Tags list for the rule. This operation is repeated for each document type.  To see which tags do not have a matching Document Map rule for a given document type, select the document type in the list. Any tag names that do not match one of the rules assigned to the selected document type, are displayed in the Unmatched Tags list. If no rules have been defined for this document type, all of the tags appear in the Unmatched Tags list.

5. Document Map Configuration

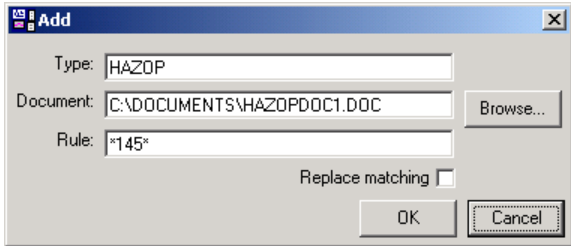
5.2. Document Map Configuration Utility

Step	Action
	TIP - Each time a list of tags is imported, any existing tag names in the Assign Document Maps tab are replaced. The new tag names are not appended to an existing list. - When no document types are selected, all of the tag names appear in the Unmatched Tags list.
3 a)	As an aid to configuring the Document Map rules, load the master list of equipment and equipment groups from ACM by clicking the Get from ACM button in the Unmatched Equipment display region. (This is the preferred method for importing the equipment and equipment groups list and it can be performed from either view.) (Optional.)
3 b)	In the Import Equipment Groups dialog box, identify the ACM control system and ACM console, and click OK.  The equipment and equipment group names are then collected directly from the specified ACM console. (NOTE: If the ACM Alarm Manager Server is not available, the Get List of Equipment Groups from ACM operation fails.)
3 c)	If ACM is unavailable or has not yet been configured, the list of equipment and equipment groups can also be collected from an Excel file by clicking the Get from Excel button in the Unmatched Equipment display region. This invokes the Get Equipment List From Excel dialog box. (This operation can be performed from either view.)

Step	Action
	 In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to launch the Excel File To Import dialog box. This dialog box may be used locate the required Excel file. Back in the Get Equipment List From Excel dialog box, identify the column in the Excel file that contains the equipment or equipment group names, by typing the column-heading name in the Column header text box. If the equipment and equipment group names are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved. Click Get List to load the equipment and equipment groups identified in the Excel workbook or click Close to cancel the load operation.
3 d)	When a list of equipment and equipment group names are loaded from either ACM or an Excel file, the Document Map Configuration utility scans the list of equipment and equipment groups names searching for those names, which match the patterns specified by the existing rules. When a match is found, the equipment or equipment group name is added to the Matched Equipment list for the rule. Again, this operation is repeated for each document type. As with tags, to see which equipment or equipment group names do not have a matching Document Map rule for a given document type, select the document type in the list. Any equipment and equipment groups names that do not match one of the rules assigned to the selected document type, are displayed in the Unmatched Equipment list. If no rules have been defined for this document type, all of the equipment and equipment groups names appear in the Unmatched Equipment list.

5. Document Map Configuration

5.2. Document Map Configuration Utility

Step	Action
	TIP - Each time a list of equipment names is imported, any existing equipment names in the Assign Document Maps tab are replaced. The new equipment names are not appended to an existing list. - When no document types are selected, all of the equipment names appear in the Unmatched Equipment list.
4	To add a new document type and, as a result, a new Document Map database file, click Add. This invokes the Add dialog box. (This operation can be performed from either view.)  In the Type text box, type the new document type name. In the Document text box, type the full path and document file name or click Browse to locate the required document file. In the Rule text box, type the rule name. Rules can contain wildcard characters (? = any single character, * = any number of characters). To have the new document type, document and rule settings, replace a previously defined document type entry, click the Replacing matching check box to select it. NOTES: - All three-text boxes must be filled in before the new document type can be added. - If an existing document type, document or Document Map rule was selected when the Add button was clicked, the Add dialog box displays the existing type, document and/or rule settings that were selected. To add a new document type, simply delete the original text box entries and enter the new settings. Click OK to accept the new document type and return to the Assign Document Maps tab, or Cancel to return to the Assign Document Maps tab without adding the new document type.

5. Document Map Configuration
5.2. Document Map Configuration Utility

Step	Action
5	To add a new document to an existing document type, select the document type and click Add. This invokes the Add dialog box, described in Step 4, with the selected document type name displayed in the Type text box. Proceed to identify the new document and an associated Document Map rule as described in Step 4.
6	To assign a rule to an existing document, select the document and click Add. This invokes the Add dialog box, described in Step 4, with the selected document and associated document type displayed in the appropriate text boxes. Proceed to identify the new Document Map rule as described in Step 4. (NOTE: Within a given document type, each Document Map rule can only be assigned to <i>one</i> document.)
7	To import a predefined set of Document Maps for multiple document types from an Excel file, click Import. This invokes the Import Documents and Rules dialog box. (This operation can be performed from either view.) In the Excel file to import text box, type the full path and file name for the Excel file or click Browse to locate the required Excel file. Identify the columns in the Excel file that contain the Document types, document names and the Document Map rule names, by typing the associated column heading name in the appropriate text box. If the Document Maps are listed in a specific worksheet in the Excel workbook, type the worksheet name in the Name of worksheet to use text box. (Optional.) NOTE: The default worksheet is the worksheet that was selected when the workbook file was last saved.

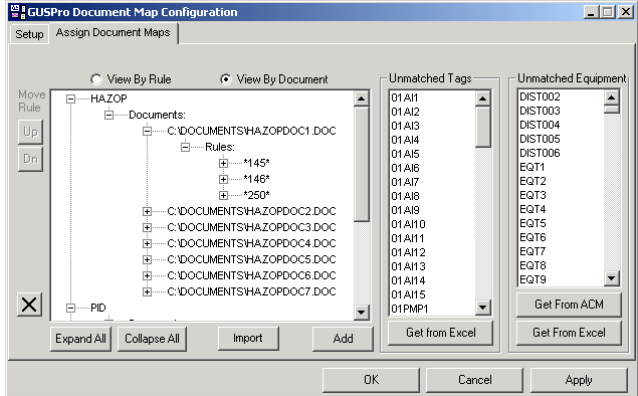
5. Document Map Configuration

5.2. Document Map Configuration Utility

Step	Action
	To have the new set of Document Map rules replace any previously defined rules with the same names, click the Replacing matching rules check box to select it. Click Get List to import the rules identified in the Excel workbook or click Close to cancel the import operation. The new rules are automatically added to the bottom of the rules list, directly above the default ("*") rule.
8	When a rule is created or a set of rules is imported, the Document Map Configuration utility scans the current lists of equipment, equipment group and tag names searching for those names, which match the patterns specified by the new rules. When a match is found, the equipment, equipment group or tag name is added to the Matched Tags or Matched Equipment list for the rule.
9	To remove an existing document type, rule or document, select the item to be removed, and click the X button. When a rule is deleted (either directly or indirectly), any equipment, equipment group or tag names that matched the deleted rule pattern are either returned to the appropriate Unmatched list or are added to another existing rule.
10	To modify an existing document type, rule or document, select the entry to be changed, pause and then single-left-click the text. When the edit box is displayed, make the desired changes and press ENTER. If a rule is changed, the Document Map Configuration utility rescans the current list of equipment, equipment group and tag names. The following results are possible: - Any equipment, equipment group or tag names that no longer match the modified rule pattern are removed from the Matched Equipment or Matched Tags list for the rule. These equipment, equipment group or tag names either are returned to the appropriate Unmatched list or are added to another existing rule for the same document type. - Any equipment, equipment group or tag names in the Unmatched lists that now match the new rule pattern are added to the Matched Equipment or Matched Tags list for the rule.

5. Document Map Configuration

5.2. Document Map Configuration Utility

Step	Action
11	To change the order in which the rules are listed, ensure that the View by Rule option is selected. (This operation cannot be performed from the View by Document view.) Select a rule that needs to be reordered and uses the Up and Dn Move Rule buttons to move it. Each time the rule order is changed, the Document Map Configuration utility rescans the list of equipment, equipment group or tag names. This may cause some of equipment, equipment group or tag names to change the rule that they are matched with.
12	The View by Document view can be used to carry out all of the configuration tasks described in Steps 2 through 10. (NOTE: The rule order cannot be changed from this view.)  To see the details for each document, double-click the document name in the list or use the Expand All and Collapse All buttons.
13	For each document type, ensure that the appropriate equipment, equipment groups and tags are covered by the required Document Map rule.

5. Document Map Configuration

5.2. Document Map Configuration Utility

Replicate the Document Map Configuration Files

Once the Document Map Configuration utility tabs have been filled in, the Document Map database files and the updated GUSPro.ini file, must be manually replicated to the appropriate computers in the system.

Step	Action
1	In the Document Map Configuration utility, click OK to close the utility and return to the GUSPro Configuration utility display. This causes a Document Map readme file (GUSProDocumentMapConfigurationReadme.txt) to be created (or updated) in the %GUSProDir% directory. If the Document Map database file storage location was changed in the Setup tab, closing the Document Map Configuration utility also causes the GUSPro.ini file to be updated. (The location of the GUSPro.ini file is determined by the %GUSProDir% environment variable.)
2	Review the contents of the GUSProDocumentMapConfigurationReadme.txt file and carry out the instructions provided as to where the Document Map Database files and GUSPro.ini file need to be copied.



ATTENTION

1. It is important that the file locations specified during Document Map configuration, be the same on all of the nodes that is accessing the Document Map database files and running the documents identified in the database.
2. If a previously defined document type is deleted in the Document Map Configuration utility, it is up to the implementer to delete the corresponding Document Map database files from all of the nodes that runs ProMenus. Failure to do so does not cause any operational problems; the files are simply not used.



TIP

%GUSProDir% is a system environment variable, which contains the name of the parent installation directory for the ProMenu and ProTrend software. If the default install directory was used, the GUSProDir directory is C:\Program Files\Honeywell\GUSPro.

6. Console Configuration

6.1 Background

GUSPro Consoles

A *GUSPro* console is a virtual console that consists of a group of nodes. In general, each *GUSPro* console includes the nodes in the operator view and the GUS stations that are part of a single TPN console, along with any supporting nodes.

Each *GUSPro* console is matched with a corresponding ACM console. This makes it possible for the equipment and equipment groups that are configured in ACM, to be used to carry out *GUSPro* configuration. UserAlert also takes advantage of the ACM console defined in the local *GUSPro* console to carry out various *GUSPro*-specific tasks.



ATTENTION

If changes are made to the ACM console database, then the *GUSPro* console must be updated to match.

See the [Configure the ACM Console Settings](#) section.

Console Databases

As part of Console configuration every application, document and display that can be launched from a ProMenu must be identified. The nodes where each application, document and display is permitted to run is also configured. This information is stored in the Console database

The Console databases are used when a request is made to invoke an application from a ProMenu graphic. When this happens, the local Console database file is checked to determine whether the application is allowed to run on the current node. If the application is not configured for the current node, the list of nodes in the Console database is searched (starting at the top of the list) and the application is launched on the first node that allows it.

6. Console Configuration

6.1. Background

The Console databases are flat text files, which use the following naming convention:

`ConsoleDefinition_<ConsoleName>.dat`

For example, if a console is named Console1, its Console database file is named "ConsoleDefinition_Console1.dat". Console Definitions files can be copied and renamed using the standard Explorer file operations, but they must be edited only through the Console Configuration Utility.

The console file must be replicated to the following directory on each node in the console: C:\HWIAC\GUSPro\



ATTENTION

One Console database file must exist for each console and must be copied onto every member node in the console. On a given node, only one Console database file must exist.

If more than one Console database file exists, the ProMenu application assumes that the node belongs to whichever Console database file that it finds first.



TIP

An implementer can create multiple console database files to configure different consoles and export them to other nodes. When configuration is complete, such local copies of "foreign" console database files must be renamed with a prefix other than "ConsoleDefinition_".

In a machine with a multi-console definition file, changes made to the following tabs are applied all the console definition files available in that node

1. Custom Applications
 2. Document Maps
 3. Display definitions
 4. ProMenu Naming Convention
-

The Console databases must only be modified using the Console Configuration utility.

Application Assignment

When a new GUSPro console is defined using the Console Configuration utility, the utility automatically assigns every identified application, document and display to every node that is a member of the console. This is done whether or not the application is actually present on each node. The implementer is responsible for ensuring that only those applications that are installed on the member nodes are configured for use on the appropriate nodes in the Console database.

The set of applications that are assigned to the nodes consists of a default set of applications, as well as the applications identified in the Console Configuration utility. The following default applications are automatically set up in each console:

- ACM Alarm Enforcer Client
- ACM Alarm Help
- UserAlert Monitor Client
- Event Analyst Client
- ProTrend Tool

All defined applications are by default allowed on all nodes in every GUSPro console identified. If one or more of the default applications are not available in the GUSPro console, the implementer is responsible for removing them from the Console database file.

The Console Configuration utility must be used to define any other applications which are available for use in the console and that may be launched by the ProMenu application.

Required logons for launching applications



TIP

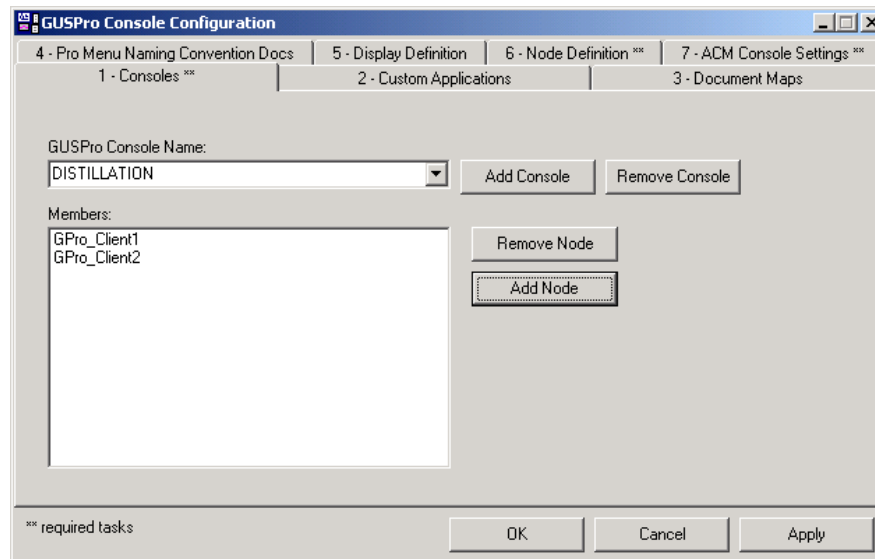
When launching applications on a remote computer using ProMenu, you must log on to both machines (the remote machine and the launching (ProMenu) machine) with the same username/password.

6.2 Console Configuration Utility

Purpose

The Console Configuration utility (Figure 15) is used to configure the Console database files, to identify the nodes in each GUSPro console and to map the GUSPro consoles to the ACM consoles.

Figure 15 Console Configuration Utility



TIP

On GUS nodes, SafeView must also be configured to display the applications or graphics that are defined in this Console Configuration utility, in the desired window.

Before You Begin

Before beginning the console configuration task, determine the computer names or IP addresses for the nodes in the GUSPro consoles.

Access

To access the Console Configuration utility, be sure that you are logged on using an account with local administrator privileges. Launch the GUSPro Configuration utility from the Windows **Start** menu by selecting **Programs > Honeywell OPS > GUSPro > GUSPro Configuration**. In the GUSPro Configuration utility, click the **Console Configuration** button.

Functions and Features

The Console Configuration utility is comprised of a series of tabbed displays. When performing the console configuration tasks, proceed from tab to tab in the order shown. Some tabs are required and some tabs are optional.

Required tabs (indicated with two asterisks **) must be completed in the order that the tabs appear. Similarly, optional tabs that are filled in must also be completed in the order in which they appear.

Duplicate entries are not allowed in any of the Console Configuration utility tabs. If an attempt is made to add a duplicate entry, a message box is displayed and the duplicate entry is ignored.

The **OK**, **Cancel** and **Apply** buttons are common to all tabs and provide the following functions:

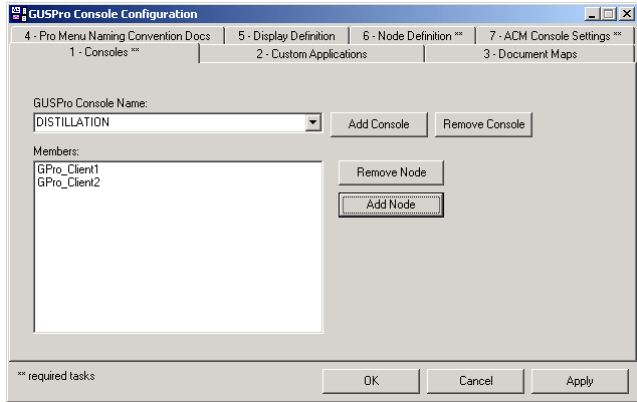
Button	Function
OK	Creates or updates the Console database files and closes the Configuration utility.
Cancel	Closes the Configuration utility without applying changes since the last apply.
Apply	Creates or updates the Console database files without closing the Configuration utility.

6. Console Configuration

6.2. Console Configuration Utility

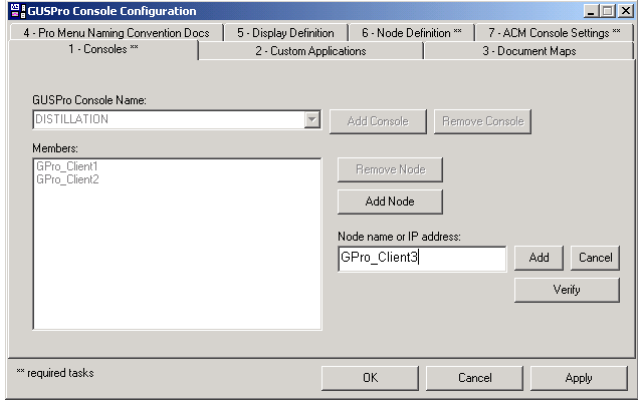
Define the Consoles and Associated Nodes

The first console configuration task is to define the set of consoles for the system and to identify the nodes, which belong to each console. The Consoles tab is used for this purpose. Upon completion of the console configuration tasks, a separate Console database file is generated for each console.

Step	Action
1	To add a new GUSPro console, open the Consoles tab and click Add Console. In the blank field in the GUSPro Console Name list box, type the console name and press ENTER. 
	TIP To simplify the configuration process, it is recommended that each GUSPro console be given the same name as the corresponding ACM console.
	ATTENTION - A GUSPro console name cannot contain a backslash (\). - If more than one console is being defined in a single session, the settings on the Custom Applications, Document Maps, Display Definitions, and ProMenu Naming Convention Docs tabs applies to all consoles.
2	When a console name is selected in the GUSPro Console Name list box, the names of the nodes currently assigned to that console appear in the Members list. To add a new member to the list, click Add Node.

6. Console Configuration

6.2. Console Configuration Utility

Step	Action
3	In the Node name or IP address text box that is displayed, identify the node and click Add.  NOTE: Duplicate node names are not allowed. To cancel the new node entry operation, click Cancel.
4	Prior to adding the node to the console members list, the Verify feature may be used to confirm that node identified in the Node name or IP address text box exists on the network. When the Verify button is clicked, a message is displayed beside the Verify button, indicating success or failure. The Verify check fails, if the node is offline, if the network isn't working or if the node doesn't exist. NOTE: Nodes may be added to a console even if the Verify check is omitted or fails.
5	To remove an existing console, select the console name in the GUSPro Console Name list box and click Remove Console.
6	To delete a node from a given console, select the console name in the GUSPro Console Name list box. In the Members list, select the node to be deleted and click Remove Node or press DELETE.
ATTENTION If a previously defined GUSPro console is deleted in the Consoles tab, it is up to the implementer to delete the corresponding Console database files from all of the nodes that will run ProMenus.	

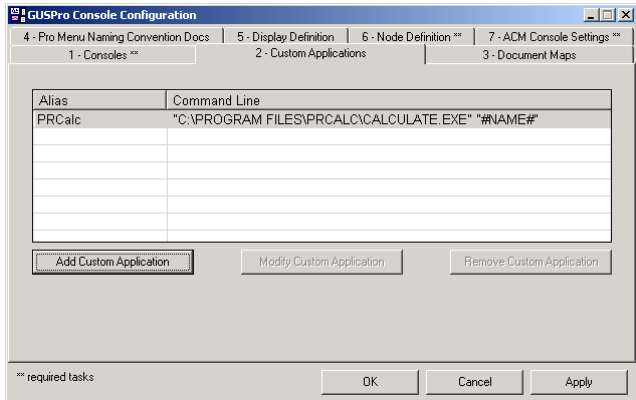
6. Console Configuration

6.2. Console Configuration Utility

Define the Custom Applications

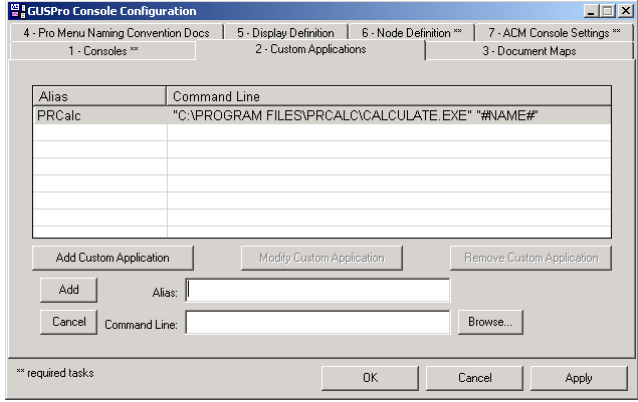
The second console configuration task is to define custom applications, which are available to all consoles. This optional operation is performed using the Custom Applications tab.

Custom applications may be launched by the ProMenu application. They are defined with an alias and a command line. The command line can contain the string #NAME# anywhere in the command. This string is replaced during operations by the ProMenu application, with a string that is passed from the calling display. This string is often a tag or equipment name.

Step	Action
1	To add a new custom application, open the Custom Applications tab and click Add Custom Application. 

6. Console Configuration

6.2. Console Configuration Utility

Step	Action
2	In the Alias text box that is displayed, type the alias which is to be used by ProMenu to identify the application. 
3	In the Command Line text box, enter the complete path string followed by the program name, or click Browse to locate the program. For command lines which include command line parameters (for example, #NAME#), be sure to include quotation marks around the command line as shown in the previous figure.
4	Once the alias and command line have been entered, click Add. To cancel the new custom application entry operation, click Cancel.
5	To remove an existing custom application, select the row containing the application alias and command line, and click Remove Custom Application or press DELETE.
6	To modify an existing custom application entry, select the row containing the application alias and command line, and click Modify Custom Application. In the Alias and Command Line text boxes that are displayed, make the desired changes and click OK. To cancel the custom application modification operation, click Cancel.
7	Ensure that every custom application that can be used in one or more of the GUSPro consoles is defined here.

6. Console Configuration

6.2. Console Configuration Utility

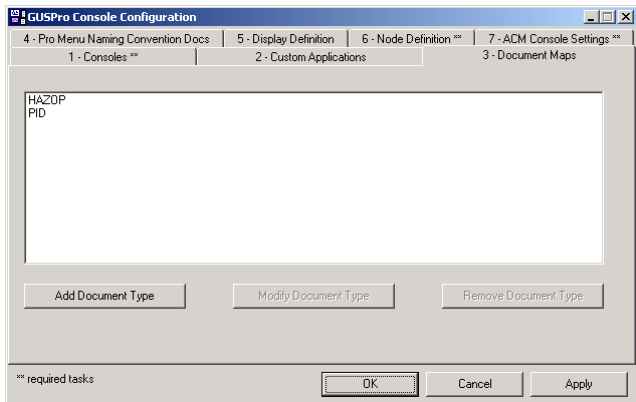
Define the Document Map Types

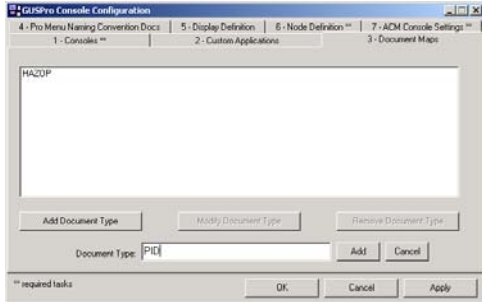
The third console configuration task is to define the Document Maps that are available to all consoles, if Document Maps are to be used. This optional operation is performed using the Document Maps tab.



TIP

Remember a separate Document Map is created for each document type using the Document Map Configuration utility, described in the Document Map Configuration Utility section. This tab is used to identify the type of document contained in each Document Map.

Step	Action
1	To add a new document type, open the Document Maps tab and click Add Document Type. 

Step	Action
2	In the Document Type text box that is displayed, type the document type name, which is used to identify the Document Map, and click Add.  To cancel the new document type entry operation, click Cancel.
3	To remove an existing document type, select the document type, and click Remove Document Type or press DELETE.
4	To edit an existing document type, select the document type name, and click Modify Document Type. In the Document Type text box that is displayed, make the desired changes and click OK. To cancel the document type modification operation, click Cancel.
5	Ensure that every document type that requires a Document Map and that can be used in one or more of the GUSPro consoles are identified here.



ATTENTION

If a new document type is added in the Document Maps tab for which there is no corresponding Document Map database file, be sure to return to the Document Map Configuration utility to create the required database file. (See the Document Map Configuration Utility section for details.)

Similarly, if an existing document type is modified in the Document Maps tab, then the associated Document Map database file must be modified and renamed accordingly using the Document Map Configuration utility.

If an existing document type is deleted from the Document Maps tab because it is no longer required, then the associated Document Map database file must be deleted from all the nodes it is currently stored on.

6. Console Configuration

6.2. Console Configuration Utility

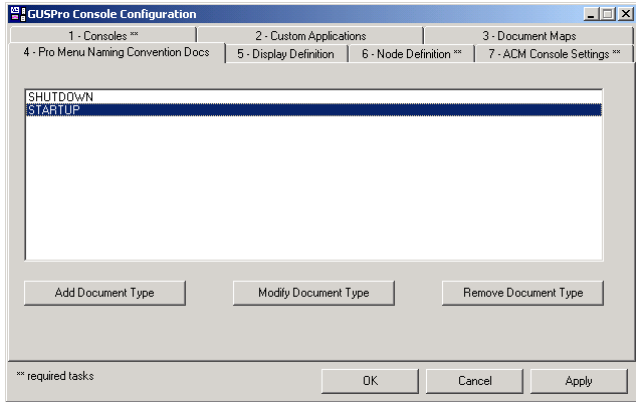
Define the ProMenu Document Types

The fourth console configuration task is to identify any document types that follow the predefined GUSPro naming convention of

`<prefixstring>_<ContextName>.<extension>`

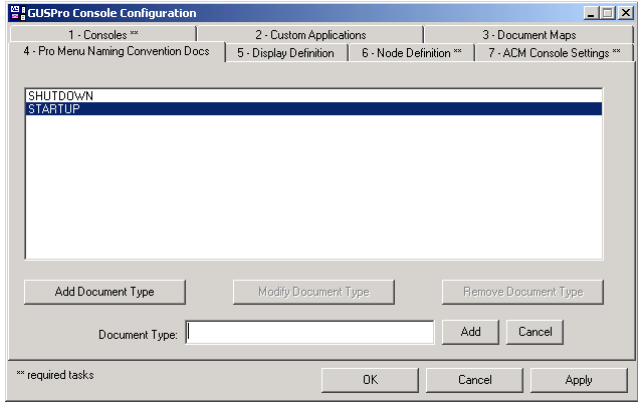
The prefix string is the actual document type. The context name is either the tag or equipment name. The prefix string for each document type is what gets defined in this tab. The document types identified are available to all consoles.

This optional operation is performed using the ProMenu Naming Convention Docs tab.

Step	Action
1	To add a prefix string for a new document type, open the ProMenu Naming Convention Docs tab and click Add Document Type. 

6. Console Configuration

6.2. Console Configuration Utility

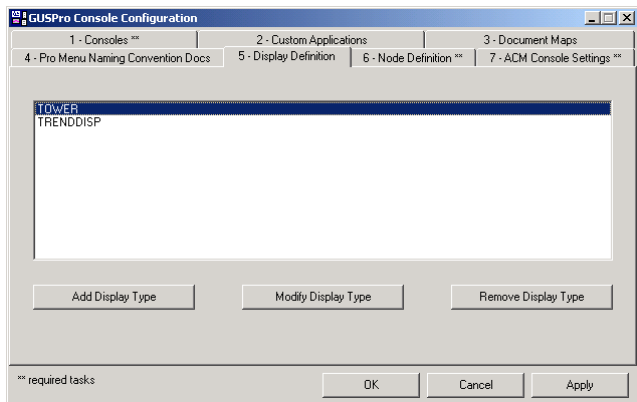
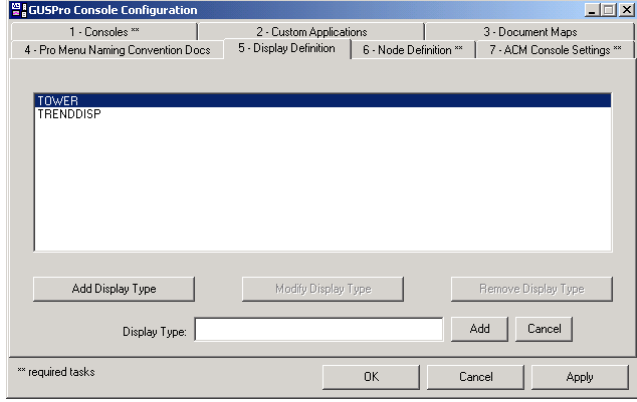
Step	Action
2	In the Document Type text box that is displayed, type the prefix string text for the document type and click Add.  To cancel the new prefix string entry operation, click Cancel.
3	To remove an existing prefix string, select the prefix string from the list, and click Remove Document Type or press DELETE.
4	To edit an existing prefix string, select the prefix string, and click Modify Document Type. In the Document Type text box that is displayed, make the desired changes and click OK. To cancel the prefix string modification operation, click Cancel.
5	Ensure that every type of ProMenu document that can be used in one or more of the GUSPro consoles is identified here.

6. Console Configuration

6.2. Console Configuration Utility

Define the GUS Graphic Types

The fifth configuration task is to define any specialized graphic display types that are to be called from the ProMenu. This optional operation is performed using the Display Definition tab.

Step	Action
1	To add a new graphic display type, open the Display Definition tab and click Add Display Type. 
2	In the Display Type text box that is displayed, type the display type name and click Add.  To cancel the new display type entry operation, click Cancel.

Step	Action
3	To remove an existing display type, select the display type name from the list, and click Remove Display Type or press DELETE.
4	To edit an existing display type, select the display type name, and click Modify Display Type . In the Display Type text box that is displayed, make the desired changes and click OK . To cancel the display type modification operation, click Cancel .
5	Ensure that every specialized graphic display type that is to be invoked by a ProMenu in one or more of the GUSPro consoles are identified here.



TIP

The display type identified in this tab is not the same as the GUS graphic category property.

Configure the Nodes That Each Application Can Run On

The sixth configuration task is to define which nodes are expected to run the applications, documents and displays defined on the previous tabs. It also uses to define which nodes are expected to run the default applications (for example: ACM Alarm Enforcer and UserAlert). This mandatory operation is performed using the Node Definition tab.

By default all applications are allowed to run on all nodes.

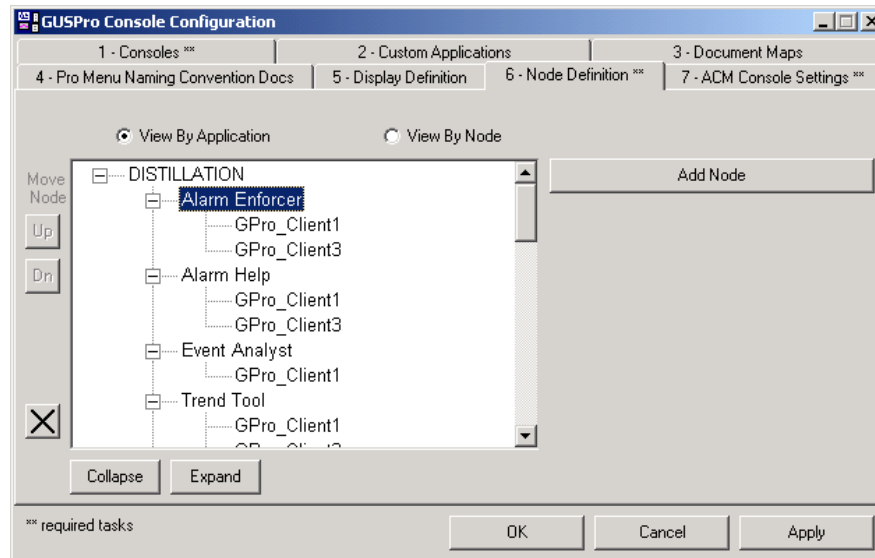
Node Definition Tab Description

There are two views for the Node Definition tab. Consoles can either be viewed by application or by node. When the **View by Application** option is selected, the applications defined in the Custom Applications tab, Document Maps tab, ProMenu Naming Convention Docs tab and Display Definition tab, are displayed in alphabetical order, along with nodes assigned to each. The set of applications displayed also includes the default applications for Alarm Configuration Manager, Event Analyst, UserAlert and ProTrend Tool.

6. Console Configuration

6.2. Console Configuration Utility

Figure 16 Node Definition Tab - View by Application



This view may be used to add or remove the nodes assigned to a given application and to change the order in which the nodes are assigned to each application. The node assignment order is important because the node order determines which node is found first in the search path.

Applications that are not installed on any of the nodes in a given console, may also be deleted from the console.

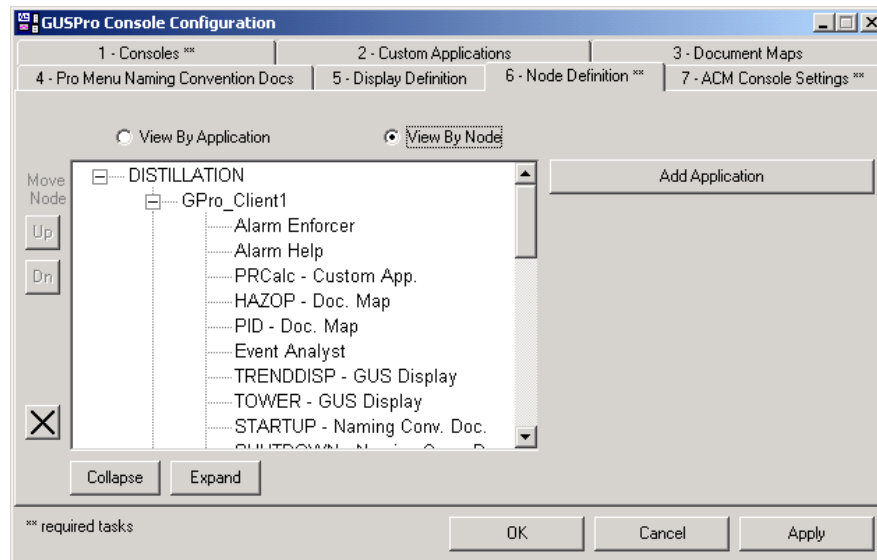


TIP

When an application is requested from a ProMenu, the application is invoked on the local node if it is allowed, regardless of the search order.

When the **View by Node** option is selected, the nodes defined in the Console tab for each console, are displayed in alphabetical order, along with applications assigned to each.

Figure 17 Node Definition Tab - View by Node



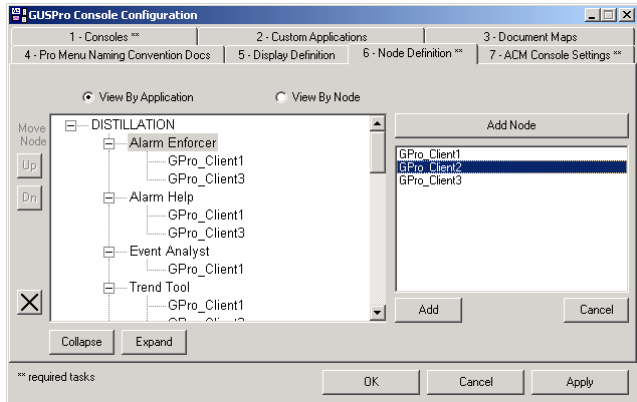
This view may be used to add or remove the applications that are assigned to each node in the console. This view cannot be used to change the node order in the application search path.

Changes made in the View by Node view are automatically reflected in the View by Application view, and vice versa.

6. Console Configuration

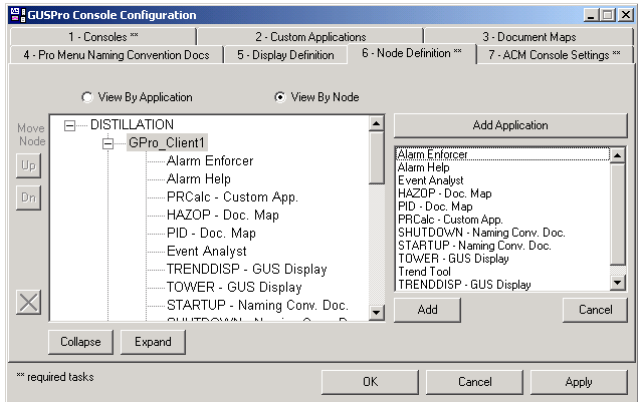
6.2. Console Configuration Utility

Configuration Procedure

Step	Action
1	Open the Node Definition tab, to show the latest Console configuration settings. By default all of the applications, documents and displays are allowed to run on all of the nodes defined as members of each console. To see the details for each console, application, document, display or node, double-click the name in the list or use the Expand and Collapse buttons.
2	To remove a node from an application, select the View by Application view. Locate the desired application and select the node to be deleted from the application's node list. Click the X button. Similarly, to remove an application from a console, select the appropriate application in the console list and click the X button.
3	To remove an application assigned to a given node, select the View by Node view. Locate the desired node and select the application to be deleted from the node's application list. Click the X button
4	To add a node to an application, select the View by Application view. Select the desired application and click Add Node.  In the list box displayed directly below the Add Node button, select the node or nodes to be added and click Add. The Add Node list box contains the nodes assigned to the console associated with the currently selected application. To cancel the add node operation, click Cancel.

6. Console Configuration

6.2. Console Configuration Utility

Step	Action
5	To add an application, document or display to a node, select the View by Node view. Select the desired node and click Add Application.  In the list box displayed directly below the Add Application button, select the application or applications to be added and click Add. The Add Application list box contains all of the applications configured in the Custom Applications tab, Document Maps tab, ProMenu Naming Convention Docs tab and Display Definition tab, along with the default Alarm Configuration Manager, Event Analyst, UserAlert and ProTrend Tool applications. To cancel the add application operation, click Cancel.
6	To change the order in which the nodes are assigned to the individual applications, select the View by Application view. In the desired application, select the node that needs to be moved and use the Up and Dn Move Node buttons to change the node position.
7	Ensure that only those applications that are actually installed on a given node are assigned to run on that node.

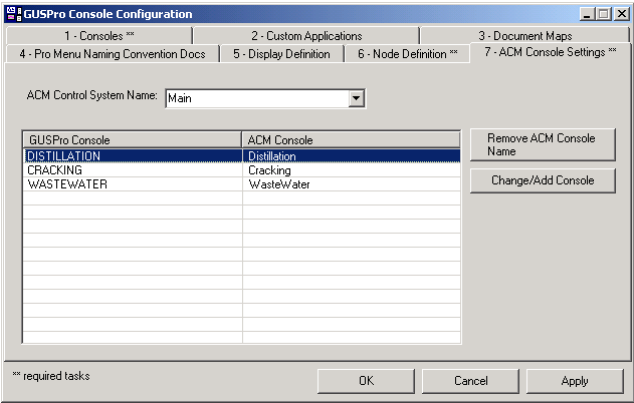
6. Console Configuration

6.2. Console Configuration Utility

Configure the ACM Console Settings

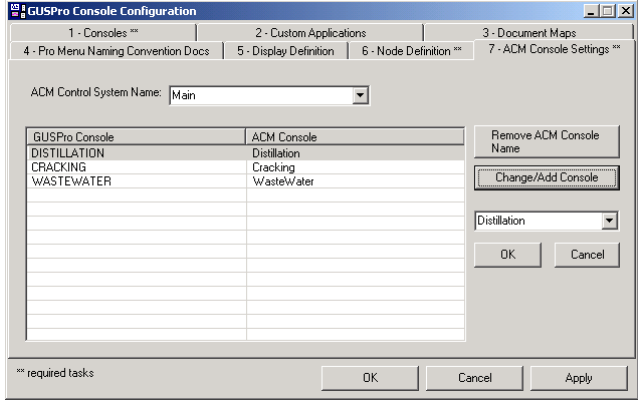
The final console configuration task is to associate the predefined ACM consoles with the GUSPro consoles. This function can only be completed on a node that has access to the ACM Alarm Manager Server that is used by the consoles. There must be a one to one relationship between GUSPro consoles and ACM consoles. Each GUSPro console can only be assigned to a single ACM console.

This mandatory operation is performed using the ACM Console Settings tab.

Step	Action
1	In the ACM Console Settings tab, select the ACM Control System Name from the list box provided.  NOTE: An ACM control system name must be selected before any ACM consoles can be selected.

6. Console Configuration

6.2. Console Configuration Utility

Step	Action
2	To assign an ACM console to a GUSPro console, select the row containing the GUSPro console of interest and click Change/Add Console.  In the list box displayed directly below the Change/Add Console button, select the ACM console to be assigned to the GUSPro console and click OK. The ACM console list box contains all of the ACM consoles configured in the specified ACM control system. To cancel the ACM console assignment operation, click Cancel.
3	To remove the ACM console assigned to a particular GUSPro console, select the row containing the GUSPro console and click Remove ACM Console Name or press DELETE.
4	To change the ACM console assigned to a given GUSPro console, select the row containing the GUSPro console and click Change/Add Console. In the list box displayed, select the new ACM console to be used and click OK.

6. Console Configuration

6.2. Console Configuration Utility

Replicate the Console Database Files

Once all of the Console Configuration utility tabs have been filled in, the Console database files must be manually replicated to the appropriate computers in the system.

Step	Action
1	In the Console Configuration utility, click OK to close the utility and return to the GUSPro Configuration utility display. This causes a Console readme file (ConsoleConfigurationReadme.txt) to be created in the %GUSProDir% directory.
2	Review the contents of the ConsoleConfigurationReadme.txt file and carry out the instructions provided as to which Console database files need to be copied where.



ATTENTION

- It is important that all GUS nodes in a given GUSPro console have the appropriate Console database file copied to the directory location specified in the GUSPro Settings display. (See the GUSPro Settings Display section for details.)
- On a given node only one Console database file must exist. If more than one Console database file exists, the ProMenu application assumes that the node belongs to the first Console database file that it finds. (NOTE: This can result in errors if the node is not defined in the file.)
- It is also important that the file locations, specified in the Custom Applications tab, be the same on all of the nodes that are running those applications.



TIP

%GUSProDir% is a system environment variable that contains the name of the parent installation directory for the ProMenu and ProTrend software. If the default install directory was used, the GUSProDir directory is C:\Program Files\Honeywell\GUSPro.

7. ProMenu Implementation

7.1 Implementation Tasks

Based on the decisions made in the ProMenu planning phase, proceed to implement the ProMenus as follows:

Task	Description	Reference
1	Implement the ProMenu Link buttons as subpictures.	ProMenu Buttons Reference section
2	Create the individual Tag and Equipment ProMenu graphics. This can be done in the GUS Display Builder, using the ProMenu template file and GUSProMenu library supplied with ProMenu. If SafeView is being used, be sure size the ProMenu graphics accordingly.	Building a ProMenu Display section SafeView Configuration section
3	Modify the target subpictures used in the existing GUS or Native Window graphics so that they can be used to launch the ProMenus. Embed the resultant subpictures in the existing displays and subpictures. Graphic scripting examples for ProMenus are provided as a starting point.	Graphic Scripting for ProMenus section

7. ProMenu Implementation

7.1. Implementation Tasks

8. ProMenu Buttons Reference

8.1 Introduction

The ProMenu button subpictures are prebuilt and delivered with the GUSPro system. They may be used to build ProMenus by the simple insertion of the button subpictures into a copy of the ProMenu template graphic (PMTemplate.pct). The implementer can modify the supplied button subpictures to create new button subpictures as required. Each ProMenu button subpicture is dynamically associated with a selected tag, equipment item or equipment group. This association is done using the ContextName and MenuType properties of the ProMenu application, Operations interface.

The button subpictures provided with ProMenu are described in the following subsections. Some buttons are appropriate for Tag ProMenus, some are appropriate for Equipment ProMenus and some are appropriate for both. All buttons inherit their font and character set from the containing graphic. By default, the action that activates each button is left button up.

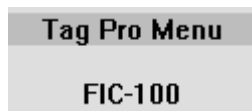
The subpictures use the following standard naming convention:

sub_PM_<specific name>.pct

The specific name describes the subpicture's function. For instance the Associated Display button is called: sub_PM_AssociatedDisplay.pct.

8.2 Common Subpictures

ProMenu Title



Basic Description

This subpicture displays a ProMenu title, the name of the selected tag, equipment item or equipment group, and the latest error (if any).

8. ProMenu Buttons Reference

8.2. Common Subpictures

If an error occurs in any of the button subpictures in the ProMenu, a message appears in the Title subpicture and an audible beep is generated. The user may clear the current error by clicking the Title subpicture, or by clicking on another button in the ProMenu without causing another error.

NOTE: The ProMenu template includes an instance of this subpicture. Normally implementation consists solely of assigning a value to the Title parameter.

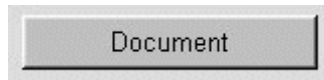
Filename

sub_PM_Title.pct

Display Parameters

Parameter	Prompt	Required Entry
Title	Please enter the title to appear on the ProMenu.	Yes

ProMenu Document Naming Convention Button



Basic Description

This button can be modified and copied to create a button for every type of document that is associated with tag and equipment names. Documents used with this button must be named using the following naming convention:

<PrefixString>_<ContextName>.<DocumentExtension>

where:

PrefixString = The document type.

ContextName = The name of the tag, equipment item or equipment group.

For a piece of equipment named TWR123, a startup procedures document written in Microsoft Word may be called "STARTUPPROC_TWR123.doc".

Filename

sub_PM_DocumentNamingConvention.pct

Display Parameters

Parameter	Prompt	Required Entry
PrefixString	Please enter the document prefix string. For example: StartupProc.	Yes
Label	Please enter the button label. Defaults to be the same as the PrefixString.	Yes The default is the value of the PrefixString parameter.
DocumentExtension	Please enter the file extension of the document.	Yes The default is .doc.
CommandLineOptions	Please enter any command line switches. If there are none then leave blank.	No Any required command line options are entered here. The default is blank.
DocumentLocation	Please enter the full path information (not including name) where the documents can be found. This can be left blank if the path is identified in the "path" system variable.	No NOTE: A backslash (\) must be included as the last character in the path name specified.



TIP

Ensure that every PrefixString parameter used, matches a prefix string defined in the Console Configuration utility - ProMenu Naming Convention Docs tab. (See the Define the ProMenu Document Types section for details.)

8. ProMenu Buttons Reference

8.2. Common Subpictures

Document Map Button



Basic Description

If predefined documents that do not use the ProMenu Document Naming Convention, need to be accessed for tags or equipment, Document Maps may be used to associate tag or equipment names with document names. See the Document Map Configuration section for a detailed description of Document Maps and how they are configured.

This button can be modified and copied to create a button for every type of document that has a Document Map configured for it.

Filename

sub_PM_DocumentMap.pct

Display Parameters

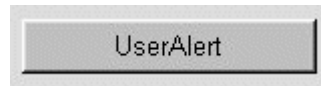
Parameter	Prompt	Required Entry
Label	Please enter the button label.	Yes The default is the value of the DocumentType parameter.
DocumentType	Please enter the Document Type.	This is the document type for the document to be invoked. Each document type corresponds to a document map as described later in this section. The document type entered must be the same as the document type configured using the Document Map Configuration utility. It is not case sensitive.



TIP

Ensure that every DocumentType parameter used, matches a Document Map document type defined in *both* the Document Map Configuration utility and the Console Configuration utility - Document Maps tab. (See the Define the Document Map Types section for details.)

UserAlert Button



Basic Description

This button is used to call the UserAlert Monitor client for an equipment group, if it is placed on an Equipment ProMenu. On a Tag ProMenu, it invokes UserAlert Monitor for the current console. (See the *UserAlert User Guide* for a description of the UserAlert application.)

Filename

sub_PM_UserAlert.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter a label for the button.	Yes The default is "UserAlert".

Custom Application Button



Basic Description

This button is used to invoke any application that a user needs that is not part of the standard ProMenu functionality.

There can be any number of these buttons on a single ProMenu graphic linking to different applications. Although any interfaces that are supported by VBScript can be put into this subpicture by modifying the script in the subpicture, only command line scripts (that have been defined in the Console configuration) are supported by the GUSPro Console configuration. This means that to have console wide invocation, only command line interfaces that are defined in the Console Configuration utility - Custom Applications tab can be used. (See the Define the Custom Applications section for details.)

8. ProMenu Buttons Reference

8.3. Equipment ProMenu Buttons

Filename

sub_PM_Custom.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the button label.	Yes
Alias	Please enter the custom application alias. It is defined during GUSPro console configuration.	Yes An alias defined in the GUSPro Console definition. This is done from the Console Configuration utility.

8.3 Equipment ProMenu Buttons

Associated Trend Button



Basic Description

This button is used to invoke the trend associated with the a specific piece of equipment. The associated trends are GUS displays that have been built by the implementer. The following naming convention must be used when configuring prebuilt trends for equipment:

TRENDDISP_<EquipName>.pct

where:

EquipName = The actual name of a piece of equipment.

Filename

sub_PM_GUSTrend.pct

Display Parameters

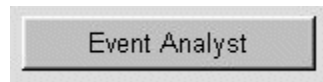
Parameter	Prompt	Required Entry
Label	Please enter the button label.	Yes The default is "Associated Trend".



TIP

In order for this button to work, TRENDISP must be defined in the console as a GUS display type. (See the Define the GUS Graphic Types section for details.)

Event Analyst Button



Basic Description

This button is used to invoke the Event Analyst client for the User File associated with a piece of equipment. To define associations between equipment and User Files, the following User File naming convention must be used:

EA_<EquipName>.txt

where:

EquipName = The actual name of a piece of equipment.

User Files within Event Analyst are used to define which tag.parameters are to appear in the Event Analyst trend. More information on creating User Files can be found in the Event Analyst user documentation.

Filename

sub_PM_EventAnalyst.pct

8. ProMenu Buttons Reference

8.3. Equipment ProMenu Buttons

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the label that appears on the button.	Yes The default is "Event Analyst".

Alarm Enforcement Button



Basic Description

This button is used to call the ACM Alarm Enforcer client. When used on an Equipment ProMenu, it passes the equipment or equipment group name to the Alarm Enforcer client, which is then invoked specifically for that equipment. (See the *ACM Alarm Enforcer User Guide* for a description of the Alarm Enforcer application and the equipment hierarchy concepts used in ACM.)



ATTENTION

It is possible, but not recommended, to use the Alarm Enforcement button on a Tag ProMenu. In this case, alarm enforcement is initiated for the entire console when the button is selected.

Filename

sub_PM_AlarmEnforcement.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the button label.	Yes The default is "Alarm Enforcement".

8.4 Tag ProMenu Buttons

Alarm Help Button



Basic Description

This button is used to call Alarm Help for a specific point.

Filename

sub_PM_AlarmHelp.pct

Display Parameters

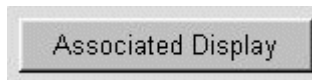
Parameter	Prompt	Required Entry
Label	Please enter the label to appear on the button.	Yes The default is "Alarm Help".



ATTENTION

If you want to select a specific parameter, then "GUSPro", "Messages" and "AHPac" must be written in the registry.

Associated Display Button



Basic Description

This button calls up the Associated display for a point. It uses the standard .ASSOCDSP parameter that is configured on LCN points.

Associated displays are only invoked on the current machine. It is assumed that SafeView is configured to handle this. Associated displays may be Native Window graphics or GUS graphics.

8. ProMenu Buttons Reference

8.4. Tag ProMenu Buttons

Filename

sub_PM_AssociatedDisplay.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the label to appear on the button.	Yes The default is "Associated Display".

Associated Equipment Menu Button



Basic Description

This button is used to invoke the ProMenu for the equipment that is associated with the selected point. This button is dependent on the equipment that has been configured in ACM.

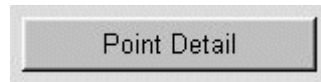
Filename

sub_PM_AssociatedEquipment.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the label to appear on the button.	Yes The default is "Associated Equipment".

Point Detail Button



Basic Description

This button is used to invoke the Native Window Point Detail display for the selected point. Native window displays are only invoked on the calling station. It is assumed that SafeView is configured to handle this.

Filename

sub_PM_Detail.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Please enter the label to appear on the button.	Yes The default is "Point Detail".

Trend Tag Button



Basic Description

This button adds the currently selected tag to the trend that is currently being displayed by the ProTrend Tool. If there is no trend displayed when this button is selected, the button starts a trend containing the selected tag.

Filename

sub_PM_TrendTag.pct

8. ProMenu Buttons Reference

8.4. Tag ProMenu Buttons

Display Parameters

Parameter	Prompt	Required Entry
Parameter	Please enter the parameter to trend.	Yes This is the parameter, for the selected tag, that is trended. The default is PV.
Label	Please enter the label for the button.	Yes The default is "Trend Tag".

9. Building a ProMenu Display

9.1 Introduction

This section explains how to build ProMenus. It assumes that implementers are familiar with basic features of the Windows Operating System and the Honeywell GUS Display Builder.

9.2 Before You Start

Determine ProMenu Size

Before you start to build ProMenus, decide on the ProMenu graphic size.

If SafeView is going to be used, be sure to review the SafeView Templates Provided section. It describes the set of SafeView templates that have been provided for use with ProMenu. The SafeView files are provided as examples only. Use as much or as little of them as is appropriate. When dealing with SafeView, it is important that you plan ahead to avoid rework.

Set Up the Registry

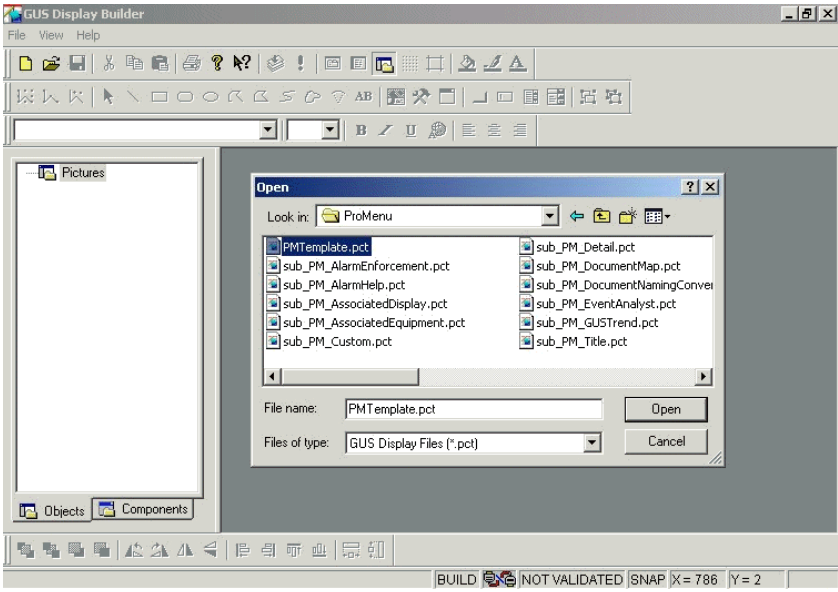
The registry is used to determine all the colors used by the ProMenu subpictures. (See the Color Registry Settings section for list of the color registry settings supplied.)

9. Building a ProMenu Display

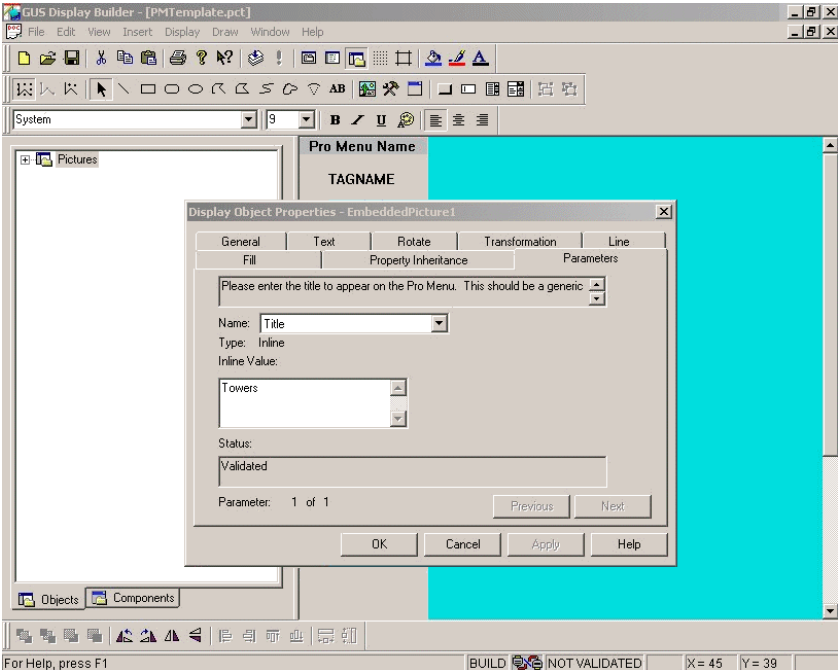
9.3. Build the ProMenu

9.3 Build the ProMenu

Open the ProMenu Template

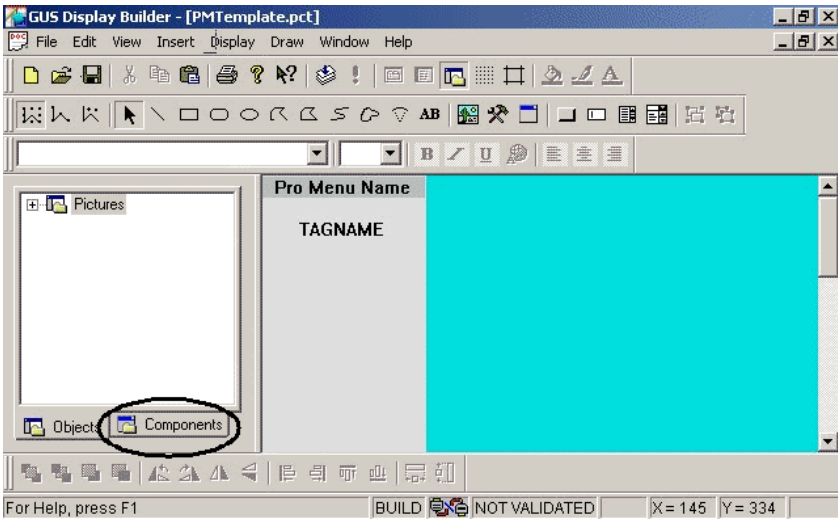
Step	Action
1	In the GUS Display Builder, open the ProMenu template file, PMTemplate.pct. This file is installed in the local \Program Files\Honeywell\GUSPro\RAC\ProMenu directory. 
2	Resize the display based on the size required for the ProMenu.

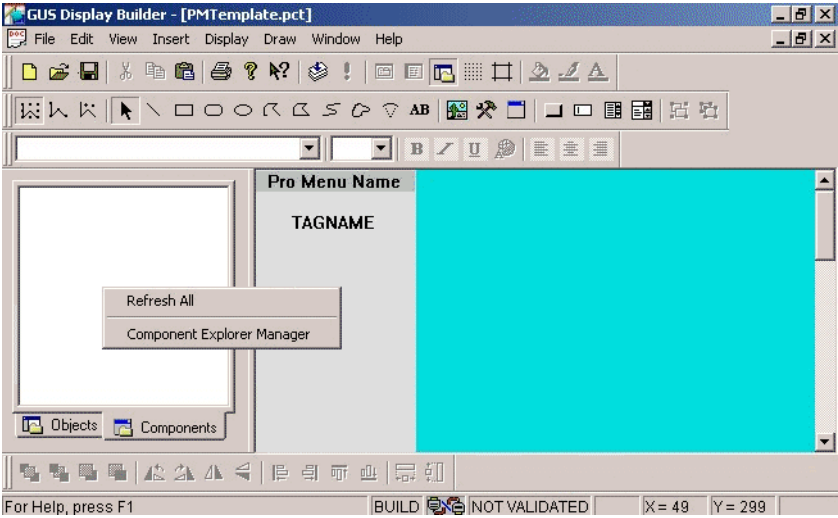
9. Building a ProMenu Display
9.3. Build the ProMenu

Step	Action
3	Double-click on the title subpicture (at the top of the ProMenu graphic) and go to the Parameters tab in the Display Object Properties dialog box.  Enter the title that is to appear at the top of the ProMenu.
4	Save the template with an appropriate name in the directory associated with Tag or Equipment ProMenus (as appropriate for this ProMenu). For example this display may be saved as "C:\Displays\ProMenus\Equipment\PMTowers.pct".

9. Building a ProMenu Display
9.3. Build the ProMenu

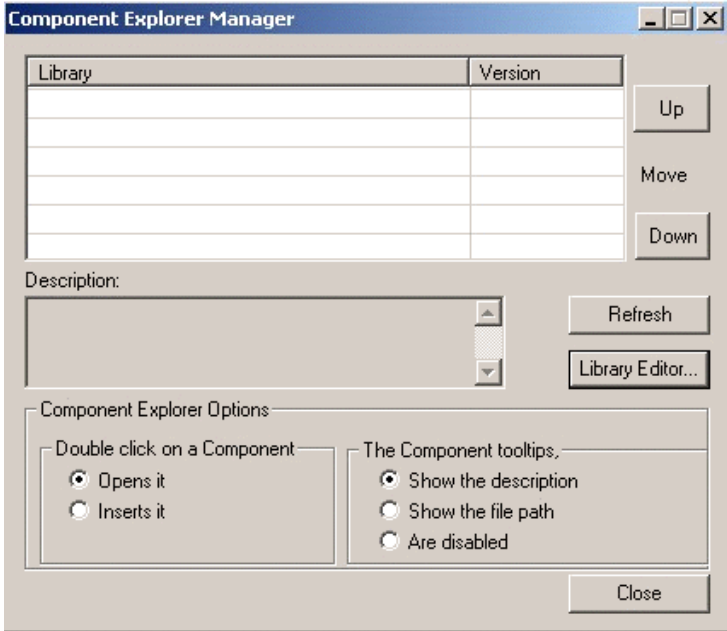
Set Up the ProMenu Component Library

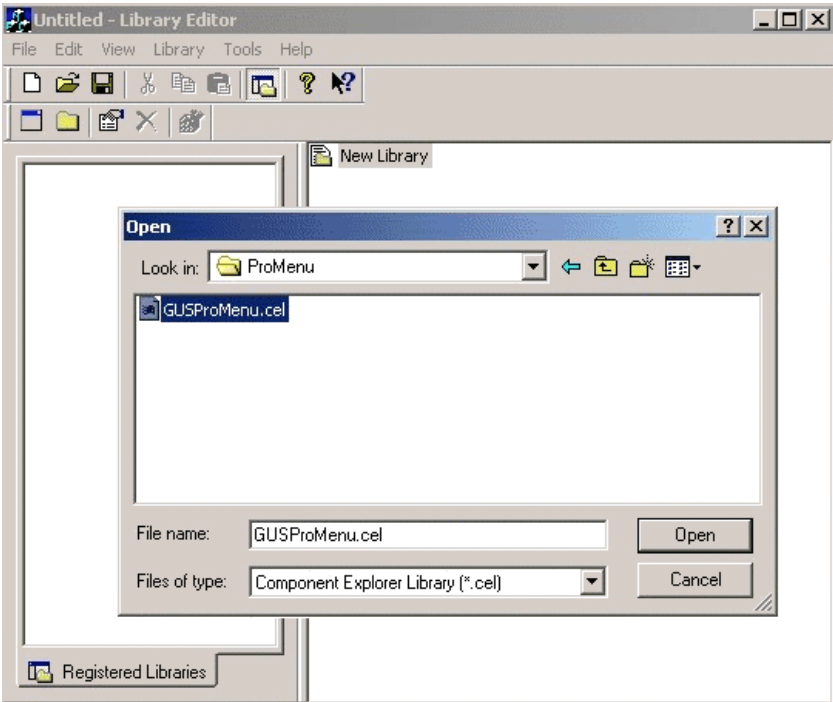
Step	Action
1	All of the ProMenu subpictures have been added to a Component Library file, which can be used for easy access to the subpictures. To reference this library file, select the Components tab at the bottom of the GUS Explorer. 

Step	Action
2	Right-click on the background of the Components Explorer view and select Component Explorer Manager from the shortcut menu. 

9. Building a ProMenu Display

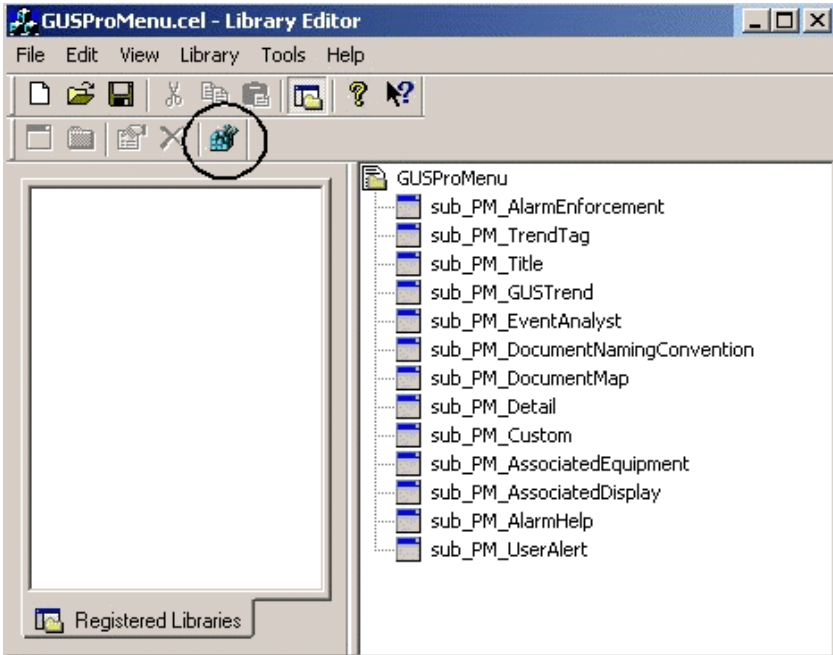
9.3. Build the ProMenu

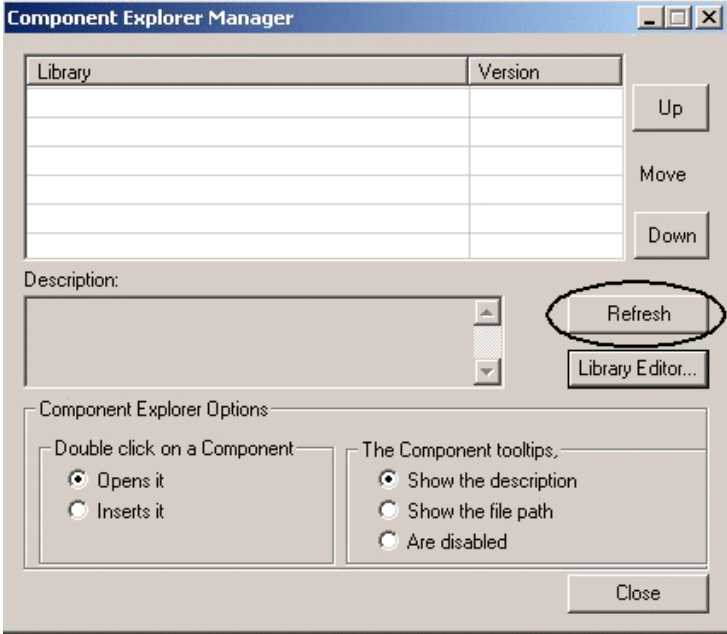
Step	Action
3	In the Component Explorer Manager, which is displayed, click the Library Editor button. This is done so that you can reference the GUSProMenu library. 

Step	Action
4	In the Library Editor, open the GUSProMenu library, GUSProMenu.cel, which is installed in the \Program Files\Honeywell\GUSPro\RAC\ProMenu directory.  The screenshot shows the 'Untitled - Library Editor' window. It has a menu bar with 'File', 'Edit', 'View', 'Library', 'Tools', and 'Help'. Below the menu bar is a toolbar with icons for file operations. The main area is divided into two panes: 'New Library' at the top and 'Registered Libraries' at the bottom. An 'Open' dialog box is overlaid on the 'Registered Libraries' pane. The 'Look in:' field shows 'ProMenu'. The file list contains 'GUSProMenu.cel', which is selected. The 'File name:' field contains 'GUSProMenu.cel'. The 'Files of type:' field is set to 'Component Explorer Library (*.cel)'. There are 'Open' and 'Cancel' buttons at the bottom right of the dialog box. At the bottom of the Library Editor window, it says 'For Help, press F1'.

9. Building a ProMenu Display

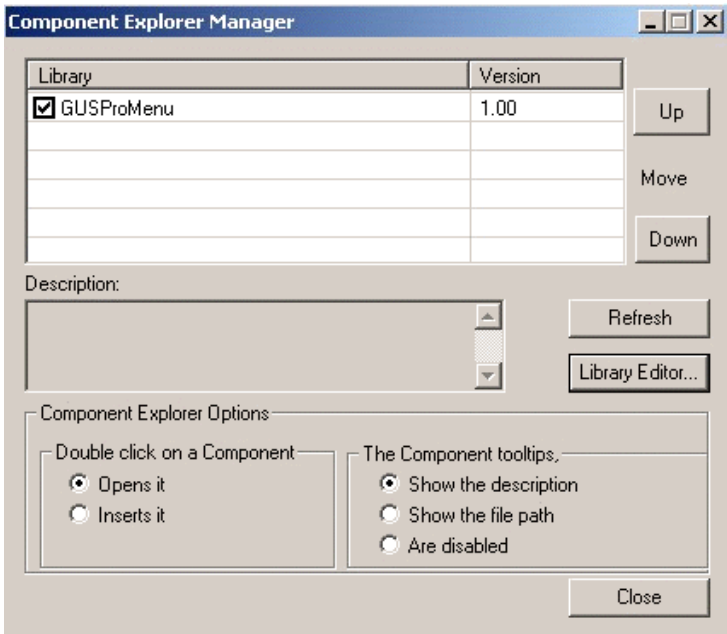
9.3. Build the ProMenu

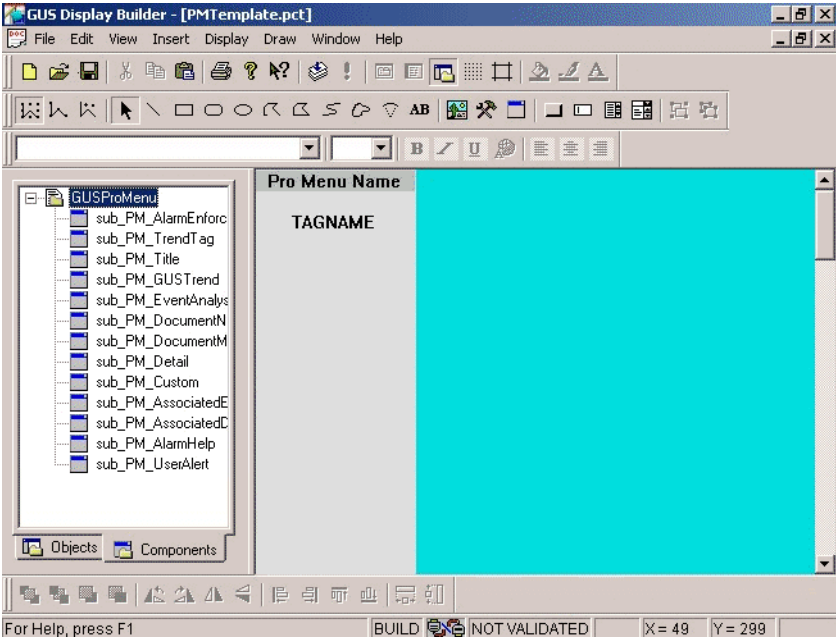
Step	Action
5	The ProMenu subpictures are displayed in the Library Editor. To register this library on the station click the register icon. 

Step	Action
6	Close the Library Editor. Back in the Component Explorer Manager, click the Refresh button to display the library.  The screenshot shows the 'Component Explorer Manager' dialog box. It has a title bar with standard window controls. Inside, there's a table with two columns: 'Library' and 'Version'. To the right of the table are three buttons: 'Up', 'Move', and 'Down'. Below the table is a 'Description:' label followed by a text area. To the right of the text area is a 'Refresh' button, which is circled in the image. Below the text area is a 'Library Editor...' button. At the bottom, there's a 'Component Explorer Options' section with two groups of radio buttons. The first group is 'Double click on a Component' with options 'Opens it' (selected) and 'Inserts it'. The second group is 'The Component tooltips,' with options 'Show the description' (selected), 'Show the file path', and 'Are disabled'. A 'Close' button is at the bottom right.

9. Building a ProMenu Display

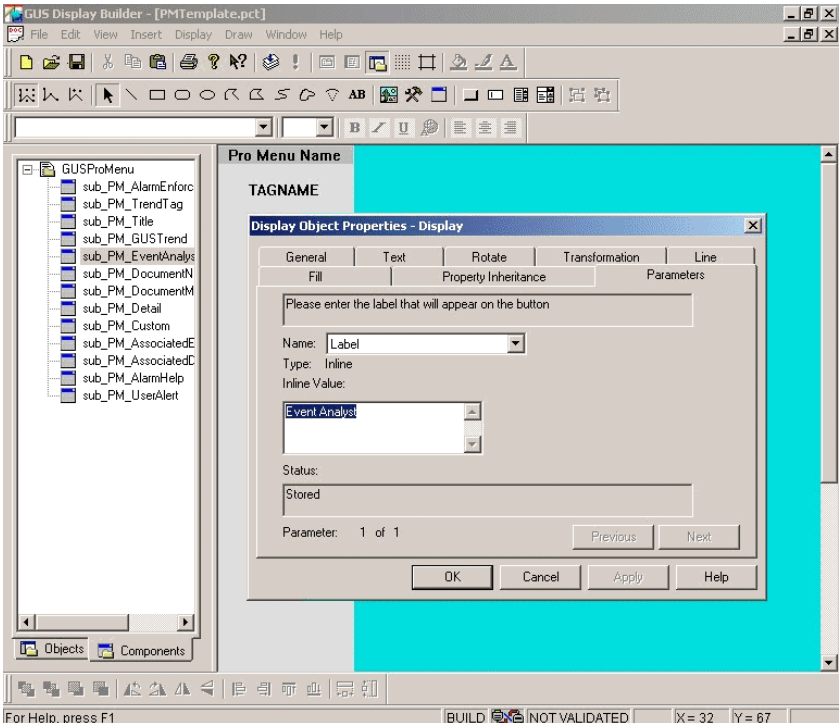
9.3. Build the ProMenu

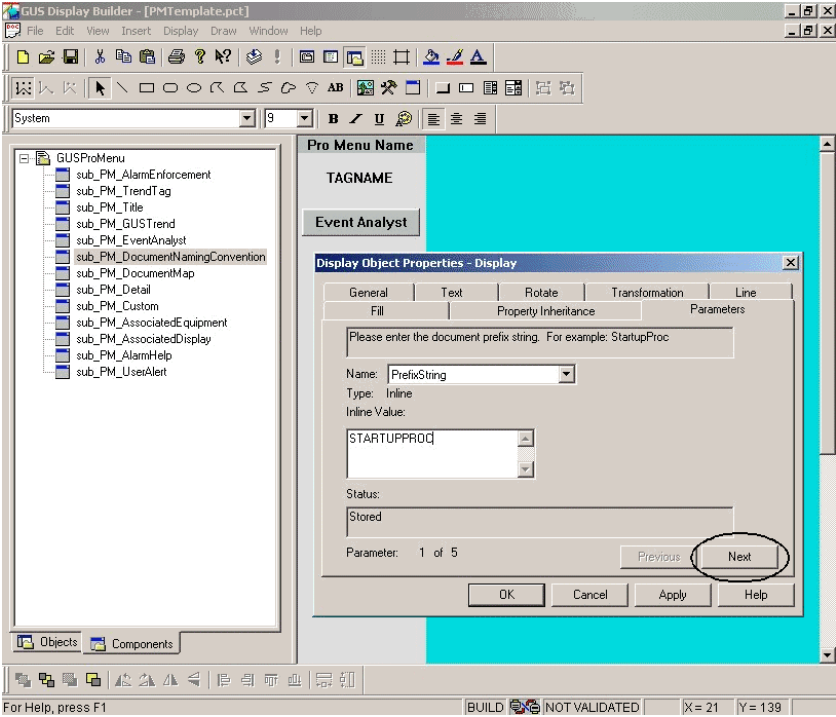
Step	Action
7	Select the GUSProMenu library (check the check box) and close the Component Explorer Manager.  NOTE: Before closing the Component Explorer Manager, the Component Explorer Options may be set.

Step	Action
8	The ProMenu subpictures now appear in the Components view of the GUS Display Builder. 

9. Building a ProMenu Display

9.3. Build the ProMenu

Step	Action
9	Insert the subpictures desired for the current ProMenu. In the following example the Event Analyst subpicture is selected. Upon insertion, you may change the label that is to appear on each subpicture.  The screenshot shows the GUS Display Builder interface. On the left, a tree view lists subpictures under 'GUSProMenu', including 'sub_PM_AlarmEnforc', 'sub_PM_TrendTag', 'sub_PM_Title', 'sub_PM_GUSTrend', 'sub_PM_EventAnalys', 'sub_PM_DocumentM', 'sub_PM_Detail', 'sub_PM_Custom', 'sub_PM_AssociatedE', 'sub_PM_AssociatedC', 'sub_PM_AlarmHelp', and 'sub_PM_UserAlert'. The 'sub_PM_EventAnalys' subpicture is selected. In the center, the 'Pro Menu Name' dialog box is open, showing the 'TAGNAME' field. The 'Display Object Properties - Display' dialog box is also open, showing the 'General' tab with the 'Name' field set to 'Label', 'Type' set to 'Inline', and 'Inline Value' set to 'Event Analyst'. The 'Status' field is set to 'Stored'. The 'Parameter' field shows '1 of 1'. The 'OK' button is highlighted.

Step	Action
10	Every subpicture has a Label parameter. Some subpictures have other parameters as well. (For example the sub_PM_DocumentNamingConvention subpicture, shown below, has multiple parameters.) If a parameter is optional, it is indicated in the description. Make sure that all required parameters are entered. To navigate between parameters click the Next button on the Parameters tab.  The screenshot shows the GUS Display Builder interface. On the left, a tree view lists subpictures under 'GUSProMenu', including 'sub_PM_AlarmEnforcement', 'sub_PM_TrendTag', 'sub_PM_Title', 'sub_PM_GUSTrend', 'sub_PM_EventAnalyst', 'sub_PM_DocumentNamingConvention', 'sub_PM_DocumentMap', 'sub_PM_Detail', 'sub_PM_Custom', 'sub_PM_AssociatedEquipment', 'sub_PM_AssociatedDisplay', 'sub_PM_AlarmHelp', and 'sub_PM_UserAlert'. The 'sub_PM_DocumentNamingConvention' subpicture is selected. On the right, the 'Pro Menu Name' dialog is open, showing the 'Event Analyst' subpicture. The 'Display Object Properties - Display' dialog is also open, showing the 'Parameters' tab. The 'Name' field is set to 'PrefixString', the 'Type' is 'Inline', and the 'Inline Value' is 'STARTUPPROC'. The 'Status' is 'Stored'. The 'Parameter' is '1 of 5'. The 'Next' button is circled.
11	Once all the desired subpictures are inserted, save the ProMenu. (See the ProMenu Buttons Reference section for a detailed description of each ProMenu button subpicture.)
	TIP If the ProMenu application is uninstalled after the GUSProMenu.cel file is used in GUS Display Builder, the Component Library file must be manually unregistered. This is done by opening the Component Explorer Manager and removing the GUSProMenu library.

9. Building a ProMenu Display

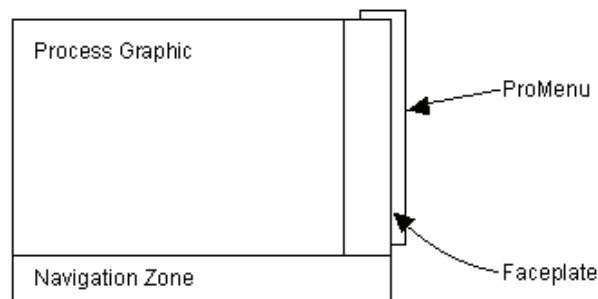
9.3. Build the ProMenu

10. SafeView Configuration

10.1 SafeView Templates Provided

Four screens are defined. These screens are arranged in different ways for different SafeView configurations – but the contents of the four screens stay the same. The SafeView templates are built for a screen resolution of 1024 x 768. Of the application windows described below the faceplate and navigation zone are locked and always visible. All other windows are global focus or otherwise as appropriate.

Screen 1

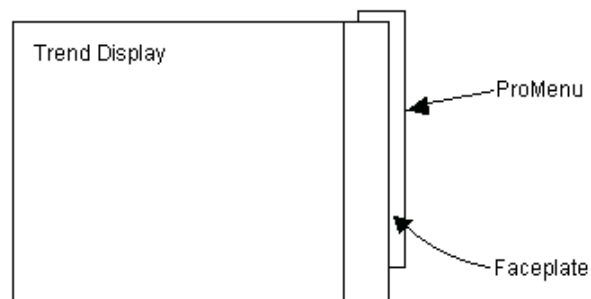


- GUS Display - Process
- GUS Display - Navigation Zone
- GUS Display - ProMenu
- GUS Display - Faceplate

10. SafeView Configuration

10.1. SafeView Templates Provided

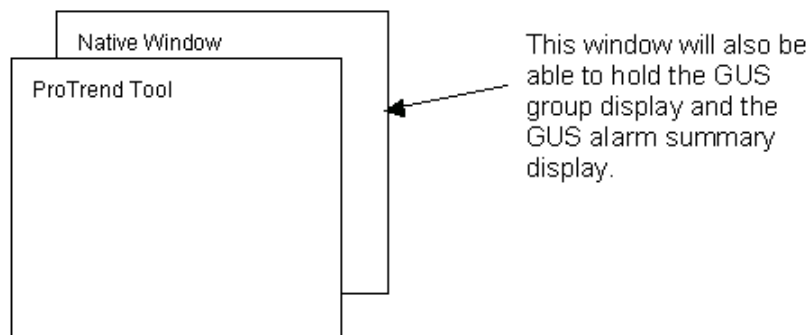
Screen 2



- GUS Display - Trend
- GUS Display - ProMenu
- GUS Display – Faceplate

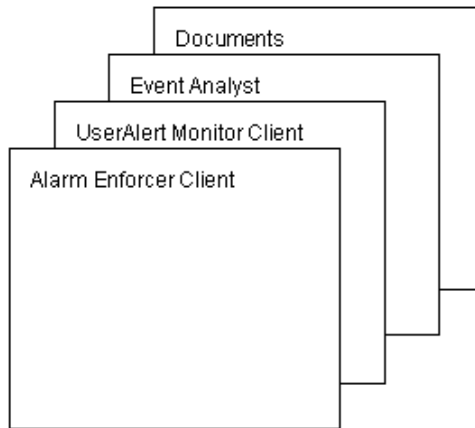
The faceplate and ProMenu only appears in this screen when the workspace is the single screen per node workspace. In all other views they do not appear on screen two because for that node they are already present on Screen 1.

Screen 3



- Native Window
- ProTrend Tool

Screen 4



- Word Documents
- ACM Enforcer Client
- UserAlert Monitor Client
- Event Analyst

Single Screen

For this SafeView configuration, the following four separate SafeView files are built for four GUS stations:

- GUSPro_SingleScreen_node1.wdl
- GUSPro_SingleScreen_node2.wdl
- GUSPro_SingleScreen_node3.wdl
- GUSPro_SingleScreen_node4.wdl

They contain the screens defined above, with node1 containing Screen1, and node2 containing Screen2, and so on. Each screen has a default window that takes up a quarter of the screen and is fully resizable and closable.

10. SafeView Configuration

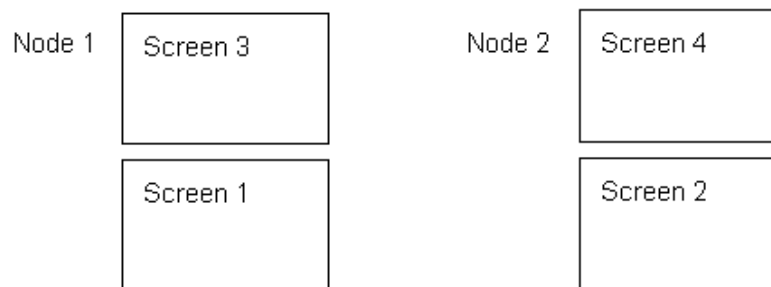
10.1. SafeView Templates Provided

Dual Screen (1 over 1)

For this SafeView configuration 2 separate SafeView files are built for 2 dual screen GUS stations. They are:

- GUSPro_DualScreen_Vertical_node1.wdl
- GUSPro_DualScreen_Vertical_node2.wdl

They are configured as follows:



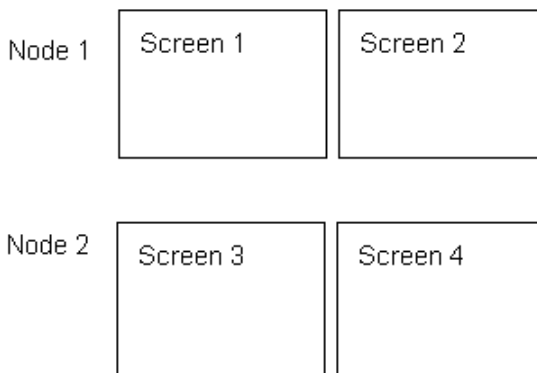
Screen 3 and Screen 4 has a default window that takes up a quarter of the screen and is fully resizable and closable.

Dual Screen Icon Console (2 Screens Side-by-Side)

This SafeView configuration contains 2 SafeView files for 2 GUS nodes.

- GUSPro_DualScreen_Horizontal_node1.wdl
- GUSPro_DualScreen_Horizontal_node2.wdl

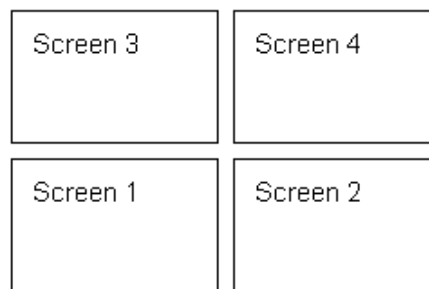
They are configured as follows:



Screen 2 and Screen 4 has a default window that takes up a quarter of the screen and is fully resizable and closable.

4-Screen Icon Console

Only one SafeView file are created for this configuration. It is called GUSPro_Icon.wdl. It is configured as follows:



Screen 2 has a default window that takes up a quarter of the screen and is fully resizable and closable.

10. SafeView Configuration

10.1. SafeView Templates Provided

11. Graphic Scripting for ProMenus

11.1 Introduction

This section describes how GUS and Native Window displays may be scripted to launch ProMenus. This scripting generally exists on process graphics. The purpose of this script is to invoke the ProMenu that is appropriate for the selected tag or equipment name.

11.2 GUS Example

An example of how to script a GUS process graphic for ProMenu invocation is provided below. You need to add a target to the display or subpicture with this script.



ATTENTION

The ProMenu object must be global in graphics that are invoking ProMenus.

Global oProMenu as Object

```
Sub OnRButtonUp()  
    On error goto ErrorHandler  
    Dim DisplayName As String  
  
    'Create ProMenu operations object  
    set oProMenu = CreateObject("GUSProMenu.Operations")  
    oProMenu.ContextName = "FIC100"  
    oProMenu.MenuType = "Tag"  
    DisplayName = oProMenu.GetProMenu  
    InvokeDisplay DisplayName  
Exit Sub  
ErrorHandler:  
Select Case err.number  
    Case 53  
        'put whatever error handling you wish to indicate  
        'that the ProMenu database wasn't found  
    Case 744  
        'put whatever error handling you wish to indicate
```

11. Graphic Scripting for ProMenus

11.3. Native Window Example

```
        'that the tag FIC100 was not found in the ProMenu
        'database
Case 429
        'put whatever error handling you wish to handle the
        'case where GUSPro has not been installed on the
        'machine.
Case else
        'Generic error handling can go here
End Select
End Sub
```

The event used (OnRButtonClick) may be changed to suit the user.

It is a possible for the ProMenu software to be installed on some GUS stations and not others. In this case the same graphics would be used on machines with ProMenus and without ProMenus. In this case the target needs to handle the error that is caused when the GUSProMenu object is not found during the CreateObject command. This is shown above with Case 429. The error raised is “ActiveX component can’t create object” – err.number = 429.

11.3 Native Window Example

To script a Native Window process graphic for ProMenu invocation, add the following line of code with the appropriate tag or equipment name to the target script in the display.

```
GP_Menu("FIC100",1)
```



ATTENTION

GUS R311.1 (or later) is required if ProMenus are going to be used in Native Window graphic displays. The MSCHEM module (version 63.3), which includes the GP_Menu actor software, is supplied with GUS R311.1.

12. ProMenu Application Programming Interface

12.1 Introduction

ProMenu Application

The ProMenu application is responsible for the operations and configuration tasks of using and building the ProMenu database and Document Map database. It consists of an ActiveX exe called GUSProMenu.exe. When called GUSProMenu.exe remains running as long as there are any references to it. A reference to it automatically exists while the ProMenuStartup application is running. Data across all references are the same.

Operations Interface

The Operations interface to GUSProMenu.exe is responsible for runtime functionality. It has several methods and properties that are defined in the following subsections. Although all properties and methods may raise errors – certain likely errors have been identified. These errors are described here for each property or method.

Accessing the Operations Class

Before using the Operations class in a GUS graphic, a reference to it must be created using the CreateObject method. Any object name may be used. In the following example (and subsequent references), the Operations class is assigned to an object called “oProMenu”.

```
Dim oProMenu as object  
set oProMenu = CreateObject("GUSProMenu.Operations")
```

12.2 Operations Interface Properties

ContextName

Syntax

ProMenuObject.ContextName

Description

This string is the name of the tag or equipment that has been selected.

12. ProMenu Application Programming Interface

12.2. Operations Interface Properties

Comments

- The ContextName property does not take any parameters.
- It is not case sensitive.
- It must be set before other methods are called that are context sensitive.
- It is persistent as long as any reference to the ProMenu application remains.
- An empty string is returned if this property is empty.
- It is a read/write property.

Example

```
oProMenu.ContextName = "FIC100"
```

Common Errors

This property does not have any specific errors defined at this time.

MenuType

Syntax

ProMenuObject.MenuType

Description

Returns the type of ProMenu being invoked. Either a string or an integer is accepted.
The accepted values are:

1 or "Tag"

2 or "Equipment"

Comments

- The MenuType property does not take any parameters.
- It is not case sensitive.
- It is a read/write property.
- It is persistent as long as any reference to the ProMenu application remains.

- It is initialized to 0.
- The return value is always the string equivalent “TAG” or “EQUIPMENT”.
- To clear this property it must be assigned the value of 0 or “”.

Example

```
oProMenu.MenuType = "Tag"
```

Common Errors

Error Number	Error Description	Possible Causes
13	Type Mismatch	Is raised if an attempt is made to write any but the accepted values to this property.

DocumentType

Syntax

ProMenuObject.DocumentType

Description

This property returns a string containing the document type for the document to be found.

Comments

- The DocumentType property does not take any parameters.
- Document types are defined during Document Map Configuration.
- This name must correspond to a Document Map database file that has been defined.
- A null string, “”, written to this property does not cause an error.
- If there is no valid document type specified this property is set to “”.
- It is not case sensitive.
- It is a read/write property.

12. ProMenu Application Programming Interface

12.2. Operations Interface Properties

Example

`oProMenu.DocumentType = "PID"`

Common Errors

Error Number	Error Description	Possible Causes
53	File not found	Is raised if the associated Document Map database file does not exist when this property is written to.

SupportsProMenus

Syntax

ProMenuObject.SupportsProMenus

Description

Returns a Boolean, which indicates if ProMenus can be launched on the current node.

Comments

- The SupportsProMenus property does not take any parameters.
- This property is true, if GUS displays (and therefore ProMenus) can be launched on the current node, and is false, if they cannot. This is determined by whether the node is a GUS or not.

Example

```
bGUSAllowed = oProMenu.SupportsProMenus
```

Common Errors

This property does not have any specific errors defined at this time.

12.3 Operations Interface Methods

GetProMenu

Syntax

ProMenuObject.GetProMenu ([mName], [mType])

Description

Returns a string with the name of the ProMenu that must be invoked for a given tag or equipment name.

NOTE: The graphic must be stored in a folder that is part of the GUS search path.

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Comments

The GetProMenu method takes the following parameters:

Parameter	Description
mName	An optional string parameter containing the tag or equipment name. If not specified, the current ContextName property is used. Setting this parameter has the side effect of setting the ContextName property.
mType	An optional string parameter containing the ProMenu type (Tag or Equipment). If not specified, the current MenuType property is used. Setting this parameter has the side effect of setting the MenuType property.

- This method returns a string containing the name of the ProMenu to be invoked.
- It is the responsibility of the calling function to invoke the graphic once the name has been returned.
- The parameters are not case sensitive.

Example

InvokeDisplay oProMenu.GetProMenu ("16FIC304", "Tag")

Common Errors

Error Number	Error Description	Possible Causes
53	File not found	Is raised if the ProMenu database file is not found.
-2146877436	The requested property value is empty.	Is raised if the ContextName or MenuType properties are empty.
744	Search Text Not Found	Is raised if there is no ProMenu that matches the context name. This only happens if a default has not been set for the MenuType.

GetDocument

Syntax

ProMenuObject.GetDocument ([context], [DocType])

Description

Returns a string with the name of the document (including path) for the tag or equipment, and document type specified in the ContextName and DocumentType properties.

Comments

The GetDocument method takes the following parameters:

Parameter	Description
context	An optional string parameter containing the tag or equipment name. If not specified, the current ContextName property is used. Setting this parameter has the side effect of setting the ContextName property.
DocType	An optional string parameter containing the document type for the document to be found. If not specified, the DocumentType property is used. Setting this parameter has the side effect of setting the DocumentType property.

- When Document Maps are built one Document Map database file is built for each type of document. Examples of this are P&ID documents and startup procedures.
- The ContextName and DocumentType properties are used to determine which document name must be returned.
- The parameters are not case sensitive.

Example

```
DocumentName = oProMenu.GetDocument ("16FIC304", "PID")
```

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Common Errors

Error Number	Error Description	Possible Causes
-2146877436	The requested property value is empty	Is raised if the ContextName or DocumentType properties are empty.
53	File not found	Is raised if the Document Map database file is not found.
744	Search Text Not Found	Is raised if there is no matching document defined for the ContextName of the given DocumentType.

GetApplicationCommand

Syntax

ProMenuObject.GetApplicationCommand(APP)

Description

Returns a string with the command for an application for a given application name. For custom applications this is a command line.

Comments

The GetApplicationCommand method takes the following parameters:

Parameter	Description
APP	A string containing the name of the application being called. This is one of the following: • ACMEnforcer • UAConfiguration • UAMonitor • EA • TrendTool • AlarmHelp • GUS_DisplayType_DisplayName – <i>DisplayType</i> must correspond to the display types defined during console configuration. <i>DisplayName</i> must be the name of a display stored in a folder in the GUS search path. • DM_DocType – This is a Document Map. <i>DocType</i> must correspond to a document map that was configured using the Document Map configuration utility. Document types are also defined during console configuration. • DNC_DocType – This is a GUSPro Naming Convention document. <i>DocType</i> must have been defined during console configuration. • Custom_StringAlias – <i>StringAlias</i> must correspond to an alias defined during console configuration. (The definition includes a string alias and a command line interface.)

- During console configuration any custom (non-standard) applications must be defined.
- This method returns the syntax needed to run the application using the RunApplication command.

NOTE: This method only returns an actual command line in the case of custom applications. In all other cases the value of APP is returned.

Example

```
oProMenu.ContextName = "16FIC304"
Command = oProMenu.GetApplicationCommand("EA")
```

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Common Errors

Error Number	Error Description	Possible Causes
53	File not found	Is raised if the Console database file is not found. (Only applies to Custom applications.)
76	Path not found	Is raised if the APP string does not match an APP defined above.
-2146877436	The requested property value is empty	Is raised if the ContextName property is empty.

GetApplicationHost

Syntax

ProMenuObject.GetApplicationHost(APP)

Description

Returns a string that identifies the computer that the required application is to be invoked on.

Comments

The GetApplicationHost method takes the following parameters:

Parameter	Description
APP	A string containing the name of the application being called. This is the same APP parameter used in the GetApplicationCommand method.

This method checks the local Console database and determines if the application is allowed to run on the current machine. If the application is allowed to run locally, the method returns the string "CURRENT". If the application is not allowed to run locally, the method searches the database and returns the name of the first machine found that allows the application to run. The Console database files are defined in the Console Configuration utility. (See the Console Configuration section for details.)

Example

```
Machine = GetApplicationHost("EA")
```

Common Errors

Error Number	Error Description	Possible Causes
53	File not found	Is raised if the Console database file can't be found.
744	Search text not found	Is raised if the application defined by APP has not been allowed on any of the nodes in the console during console configuration.

RunApplication

Syntax

ProMenuObject.RunApplication Command, Host, Context

Description

Runs the requested application.

Comments

The RunApplication method takes the following parameters:

Parameter	Description
Command	This string is the command returned from GetApplicationCommand method.
Host	A string containing the name of the computer on which the application is to be run.
Context	The string that appears in the Context parameter depends on the Command parameter.· The following table outlines the expected Context parameters for each Command string.

Command	Context	Comments
---------	---------	----------

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Command	Context	Comments
ACMEnforcer	blank	If the MenuType is "TAG".
	ContextName	If the MenuType is "EQUIPMENT".
UAConfiguration	blank	The Context parameter is not used in this case.
UAMonitor	blank	If the MenuType is "TAG".
	ContextName	If the MenuType is "EQUIPMENT".
EA	ContextName	Only works if the MenuType is "EQUIPMENT".
TrendTool	ContextName	Only works if the MenuType is "TAG".
AlarmHelp	ContextName. Parameter	Where Parameter is the parameter that must be in focus in the AlarmHelp Window.
GUS _DisplayType _DisplayName	blank	The Context parameter is not used in this case.
DM_DocType	ContextName	
DNC_DocType	The name of the document to be called.	This document name must take the following form: <PrefixString>_<ContextName>.<Document Extension> where: PrefixString = The document type. ContextName = The tag or equipment name. For a piece of equipment named TWR123, a startup procedures document written in Microsoft Word could be called: "STARTUPPROC_TWR123.doc".
Custom_String Alias	ContextName	

This method is called on the actual host machine that is to run the application.

Example

RunApplication Command, Host, context

Common Errors

Error Number	Error Description	Possible Causes
463	Class not registered on the Local Machine	Is raised if the host machine can't be found.
429	ActiveX component can't create object	Is raised if the ProMenu application, GUSProMenu.exe, is not installed on the host machine. Errors are also returned on the host machine, if there is a problem with the application being called.

QuickAppRun

Syntax

ProMenuObject.QuickAppRun APP, [Context], [DocType], [MenType], [Parameter], [DocumentName]

Description

This method encapsulates the methods needed to run an application into one method.

Comments

The QuickAppRun method takes the following parameters:

Parameter	Description
APP	A string containing the name of the application being called. This is the same APP parameter used in the GetApplicationCommand method.
Context	An optional string parameter containing the tag or equipment name. Setting this parameter has the side effect of setting the ContextName property of the object.

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Parameter	Description
DocType	An optional string parameter containing the document type for the document to be found. Setting this parameter has the side effect of setting the DocumentType property of the object.
MenType	An optional string parameter containing the ProMenu type (Tag or Equipment). Setting this parameter has the side effect of setting the MenuType property of the object.
Parameter	When the APP parameter is set to "AlarmHelp", this parameter is used to determine which parameter is in focus for the AlarmHelp window. For all other settings of APP, this parameter is ignored. If APP is set to "AlarmHelp" and this parameter is left blank, an error is raised.
DocumentName	When the APP parameter is set to DNC_DocType, this parameter is used to determine which document is to be invoked. If it is not there, an error is raised. For all other settings of APP, this parameter is ignored.

This method encapsulates the GetApplicationCommand, GetApplicationHost, and RunApplication methods.

Example

```
oProMenu.ContextName = "FIC12543"  
oProMenu.QuickAppRun"ACMManager"
```

Common Errors

Error Number	Error Description	Possible Causes
53	File not found	Is raised if the Console database file is not found. (Only applies to Custom applications.) It is also raised if the ProMenu database file can't be found.
76	Path not found	Is raised if the APP string does not match a defined APP.
-2146877436	The requested property value is empty	Is raised if the ContextName property is empty. This error is also raised if the Parameter parameter is empty for the AlarmHelp application or the DocumentName parameter is empty for DNC documents.
744	Search text not found	Is raised if the application defined by APP has not been allowed to run on any of the nodes in the console. Based on the settings in the Console database file, configured using the Console Configuration utility.
463	Class not registered on the Local Machine	Is raised if the host machine can't be found.
429	ActiveX component can't create object	Is raised if the ProMenu application is not installed on the host machine.

LaunchTagProMenu

Syntax

ProMenuObject.LaunchTagProMenu [Context]

Description

Launches the Tag ProMenu on the current machine.

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Comments

The LaunchTagProMenu method takes the following parameters:

Parameter	Description
Context	An optional string parameter containing the tag name. Setting this parameter has the side effect of setting the ContextName property of the object.

- In addition to setting the ContextName property, this method also sets the MenuType property of the object to "TAG".
- This method was created so that applications (such as UserAlert), that have no knowledge of launching GUS displays, can simply launch a Tag ProMenu for a given tag name.

Example

```
oProMenu.LaunchTagProMenu "FIC12543"
```

Common Errors

Error Number	Error Description	Possible Causes
-2146877436	The requested property value is empty	Is raised if the ContextName property is empty.
53	File not found	Is raised if the ProMenu database file is not found.
744	Search Text Not Found	Is raised if there is no ProMenu that matches the context name. This only happens if a default has not been set for the MenuType.

LaunchEquipmentProMenu

Syntax

ProMenuObject.LaunchEquipmentProMenu [Context]

Description

Launches the Equipment ProMenu on the current machine.

Comments

The LaunchEquipmentProMenu method takes the following parameters:

Parameter	Description
Context	An optional string parameter containing the equipment name. Setting this parameter has the side effect of setting the ContextName property of the object.

- In addition to setting the ContextName property, this method also sets the MenuType property of the object to "EQUIPMENT".
- This method was created so that applications (such as UserAlert), that have no knowledge of launching GUS displays, can simply launch an Equipment ProMenu for a given equipment name.

Example

```
oProMenu.LaunchEquipmentProMenu "TOWER1235"
```

12. ProMenu Application Programming Interface

12.3. Operations Interface Methods

Common Errors

Error Number	Error Description	Possible Causes
-2146877436	The requested property value is empty	Is raised if the ContextName property is empty.
53	File not found	Is raised if the ProMenu database file is not found.
744	Search Text Not Found	Is raised if there is no ProMenu that matches the context name. This only happens if a default has not been set for the MenuType.

GetACMConfiguration

Syntax

ProMenuObject.GetACMConfiguration ACMConsoleName, ACMControlSystemName

Description

Searches the current Console database file and determines the ACM console name and control system name.

Comments

The GetACMConfiguration method takes the following parameters:

Parameter	Description
ACMConsoleName [out]	This string is the ACM Console Name as defined in the Console database file.
ACMControlSystemName [out]	This string is the ACM Control System Name as defined in the Console database file.

This method is used by other applications, such as UserAlert, to determine the ACM context that they are operating under.

NOTE: Both parameters are returned in the call – not set.

Example

```
Dim sACMConsole as string
Dim sACMControlSystem as string
oProMenu.GetACMConfiguration sACMConsole, sACMControlSystem
```

Common Errors

This method does not have any specific errors defined at this time.

12.4 Example

Invoking a Document Defined in a Document Map

This example shows how a document, which has been defined as being part of a document type in a Document Map, is invoked. This typically is done within a ProMenu display. In the following it is assumed that it has an inline parameter for the document type called DocType. For this example the actual tagname is being retrieved from the registry. This script exists on the Document Map button:

```
Sub OnLButtonUp()
    On error goto ErrorHandler
    Dim oProMenu as object
    Dim sName As String
    Dim sDocument As String
    Dim sCommand As String
    Dim sHost As String
    Dim bRet As Boolean

    'Create the oProMenu object
    set oProMenu = CreateObject("GUSProMenu.Operations")
    'This assumes that the GUS Display that called this
    'ProMenu wrote the Tag or Equipment name to this
    'location in the registry
    sName = GetSetting("GUSPro", "Menu", "Name")
    'Determine what the command line is for the application.
    'PID is a document map that has been defined in the
    'console configuration and in the document map
    'configuration
    oProMenu.ContextName = sName
    sCommand = oProMenu.GetApplicationCommand("DM_PID")
```

12. ProMenu Application Programming Interface

12.4. Example

```
'Find out what node this is supposed to be called on
sHost = oProMenu.GetApplicationHost("DM_PID")
```

```
'Run the application
oProMenu.RunApplicationsCommand, sHost, sName
```

Exit Sub

ErrorHandler:

```
Select Case Err.number
Case 53
    'put whatever error handling you wish to
    'indicate that the Document map database, or
    'the console configuration file wasn't found
Case 744
    'put whatever error handling you wish to
    'indicate that the tag, sName or the document
    'type was not found in the document map
    'databases or the document type has not been
    'allowed on any machine in the console
Case 76
    'put whatever error handling you wish to
    'indicate that the document type was not found
    'in the console database
Case 463
    'put whatever error handling you wish if the
    'host machine is not found
Case 429
    'put whatever error handling you wish to handle
    'the case where GUSPro has not been installed
    'on the machine.
Case else
    'Generic error handling can go here
```

End Select

End Sub

13. Appendix

13.1 Color Registry Settings

The following registry settings are used by the ProMenus and by the ProTrend Object, and are located in “BasicScript Program Settings>GUSPro>Colors>” registry.

Registry Key	Default Color		
	Value	Color Description	RGB()
ErrorFill	33554432	Black	0, 0, 0
Text	33554432	Black	0, 0, 0
TitleBack	45857723	Gray	187, 187, 187
DisplayBack	48094685	Light Gray	221, 221, 221
ButtonNormal	46186688	Gray	192, 192, 192
ButtonSelected	48292064	Gray	224, 224, 224
ButtonText	33554432	Black	0, 0, 0
EmerAlarm	33554687	Red	255, 0, 0
ErrorBack	33554432	Black	0, 0, 0
ErrorText	50331647	White	255, 255, 255
HighAlarm	33619967	Yellow	255, 255, 0
LowAlarm	50331392	Cyan	0, 255, 255
MainBacking	46976204	Light Gray	204, 204, 204
MainText	33554432	Black	0, 0, 0
ModeHighlight	44739242	Medium Gray	170, 170, 170
OPTrace	45741567	Light Yellow	288, 245, 185
PVTrace	33587236	Dark Green	36, 128, 0
SecondaryBacking	44739242	Medium Gray	170, 170, 170

13. Appendix

13.1. Color Registry Settings

Registry Key	Default Color		
	Value	Color Description	RGB()
SecondaryText	50331647	White	255, 255, 255
Selected	39821195	Dull Green	139, 159, 95
SPTrace	46102272	Medium Blue	0, 119, 191
TrendBack	44739242	Medium Gray	170, 170, 170
DisplayBackground	48094685	Light Gray	221, 221, 221
Trace1	39288832	Dark Blue Green	0, 128, 87
Trace2	41963264	Dark Blue	0, 79, 128
Trace3	41943104	Purple	64, 0, 128
Trace4	39059584	Dark Pink	128, 0, 84
HighlightBack	50331647	White	255, 255, 255
TitleText	33554432	Black	0, 0, 0

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